TRADING FOUNDRY

COURSE PLAYBOOK · INTERMEDIATE · 10 LESSONS

The MACD *Masterclass.*

Read MACD as what it really is — speed and acceleration — and stop paying for crossovers that arrive after the move.



EDUCATION ONLY — NOT FINANCIAL ADVICE. CHARTS ARE ILLUSTRATIVE, SYNTHETIC PRICE DATA.

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What you will be able to do

- Explain exactly what the MACD line, signal line and histogram measure — and what they cannot see
- Measure the lag in any crossover and judge when it makes a trade not worth taking
- Use the MACD zero line, on your chart and the higher timeframe, to decide which direction you may trade
- Time the end of a pullback with the MACD reset and the histogram hook, confirmed by a price trigger
- Identify genuine MACD divergence and trade it only after price structure confirms
- Recognise the ranges where MACD whipsaws and switch your trend methods off in time
- Classify the market regime and choose the matching MACD rule set, with stops, targets and fixed risk
- Test a written MACD method honestly before trusting it with real money

This playbook is educational content only. It is not financial advice and nothing in it is a recommendation to buy or sell any instrument. Every chart is an illustrative training example built from synthetic price data — not a record of real market history or results. Trading foreign exchange on margin carries a high level of risk.

What MACD Actually Measures.

MODULE 1 — WHAT MACD REALLY MEASURES · LESSON 1

Read the MACD line, signal line and histogram as what they really are — a speed reading and an acceleration reading of price — rather than as buy and sell lights.

Most traders meet MACD as a pair of wiggly lines under the chart and a rule somebody posted online: buy when the blue line crosses above the orange one, sell when it crosses back. They try it, it works brilliantly on the three examples in the video, and then it quietly bleeds their account in the following month. The usual conclusion is that MACD "doesn't work".

MACD works perfectly. It does exactly what its arithmetic says it does, every single candle, without fail. The problem is that very few people are told what that arithmetic is. Once you know, the indicator stops being a set of traffic lights and becomes something far more useful: an honest measuring instrument for the speed of a trend and whether that speed is rising or falling.

This lesson builds that understanding from the ground up. There are no trade setups here yet. By the end, you should be able to look at any MACD pane and say, in plain English, what price has been doing — and, just as importantly, what MACD cannot possibly tell you.

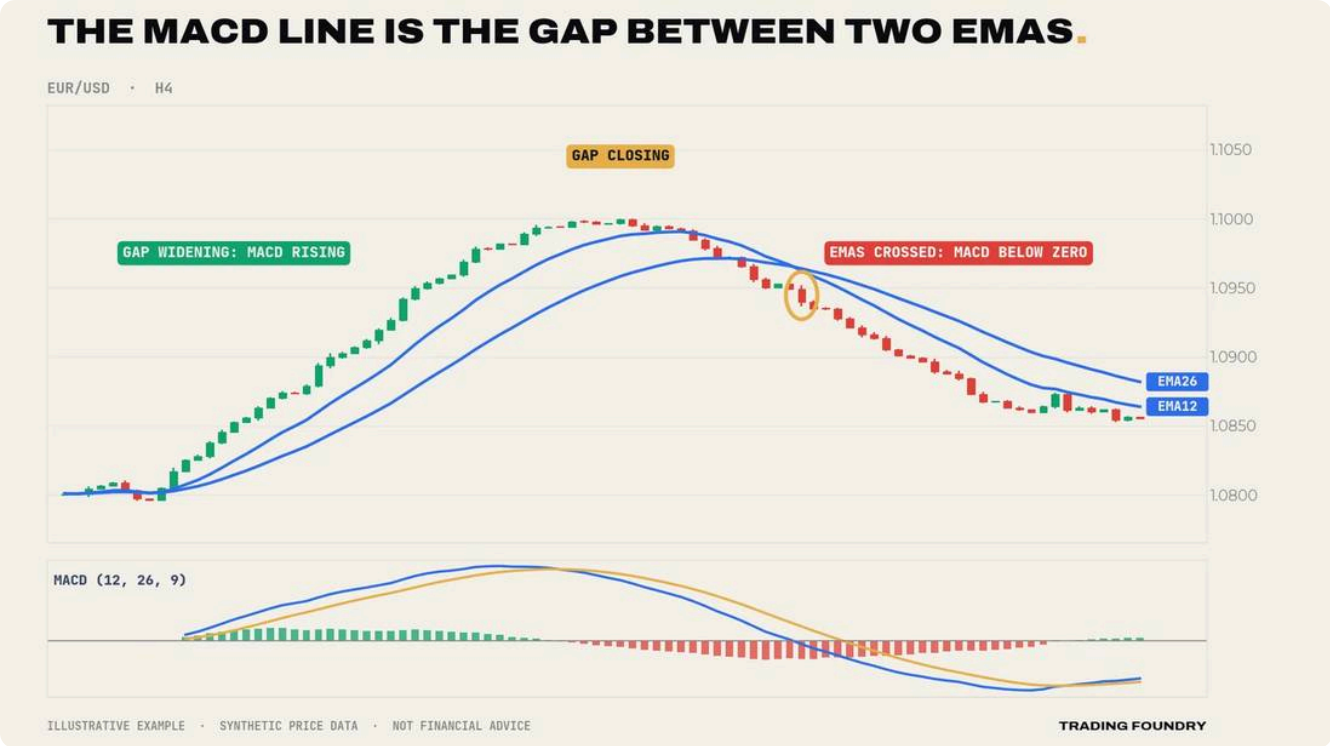
Three numbers, one idea

MACD stands for Moving Average Convergence Divergence. It was developed by Gerald Appel in the late 1970s, and the histogram was added later by Thomas Aspray. Strip away the name and it is three numbers, recalculated on every candle:

1. **The MACD line.** The 12-period exponential moving average (EMA) of the closing price, minus the 26-period EMA. That is all. When the fast average is above the slow one, the MACD line is positive. When it is below, the line is negative.
2. **The signal line.** A 9-period EMA of the MACD line itself — an average of the difference.
3. **The histogram.** The MACD line minus the signal line, drawn as bars.

The settings you see printed as "MACD (12, 26, 9)" are simply those three periods. Every MACD chart in this course uses them, because they are what almost every platform loads by default. Lesson 3 deals with whether changing them helps (mostly, it does not).

The name tells you the idea. The two averages **converge** when they move towards each other and **diverge** when they spread apart. The MACD line is a single number that records the gap between them.



The two lines on price are the 12 and 26 EMA. While the rally is strong the gap between them widens and the MACD line climbs. As price stalls the gap closes, and the moment the 12 EMA drops through the 26 EMA (circled) the MACD line crosses zero. The pane below is nothing more than that gap, plotted on its own.

Look at the chart above slowly. The MACD pane is not adding information that the price pane does not contain. It is re-plotting one feature of the price pane — the distance between two averages — so that your eye can read it without squinting at two lines that are close together.

KEY POINT

The MACD line is the 12 EMA minus the 26 EMA. Nothing more mysterious is going on.

The MACD line is a speedometer

Why would the gap between a fast and a slow average matter? Because of lag. An EMA trails behind price, and a slower EMA trails further. In a steady trend, the 12 EMA sits a little behind price and the 26 EMA sits further behind. The faster price is travelling, the further behind both averages fall, and the wider the gap between them becomes.

So the MACD line is, in effect, a **speedometer for the trend**:

- A large positive value means price has been rising quickly over the last few weeks of candles.
- A small positive value means price has been rising slowly, or has stopped rising recently.
- Zero means the fast and slow averages agree — price has, on balance, gone nowhere over the lookback.
- Negative values are the same story for falling prices.

That framing immediately clears up a common confusion. A rising MACD line does not mean price is rising. It means price is rising **faster than it was**. A falling MACD line that is still above zero does not mean price is falling. It means the uptrend is losing speed. Price can make new highs while the MACD line falls, simply because each new high arrived more slowly than the last. Lesson 6 turns that observation into a tool called divergence.



Price keeps making higher highs all the way to the right-hand side, but the steepest part of the rally is in the first third. The MACD line peaks around the same place — where price was travelling fastest — and declines while price grinds higher. A falling MACD above zero means a slowing uptrend, not a falling market.

There is a simple piece of arithmetic behind this, and it is worth knowing because it removes all the mystery. An EMA lags a steady trend by roughly half its period. The 12 EMA trails by about 5.5 candles and the 26 EMA by about 12.5. In a perfectly steady trend moving k pips per candle, the gap between them settles at roughly $7 \times k$. A pair climbing a steady 3 pips per candle will show a MACD line of about +21 pips. Speed it up to 5 pips per candle and the reading settles near +35. The MACD line really is a speed reading, measured in pips.

The signal line and histogram: acceleration

If the MACD line is speed, the histogram is **acceleration**.

The signal line is an average of the MACD line over the last nine candles. Comparing the MACD line with its own recent average tells you whether speed is currently above or below what it has been lately:

- Histogram bars above zero and growing: the trend is speeding up.
- Histogram bars above zero but shrinking: still faster than the recent average, but the increase is fading.
- Histogram crossing below zero: speed has dropped below its recent average. Something is slowing.

The histogram crossing zero is exactly the same event as the famous "MACD crosses the signal line". They are one event drawn two ways. When you hear a trader talk about a "bullish crossover", they mean the histogram has turned from red to green.



Two pushes higher, each one paid for with growing green histogram bars. During the pause the bars shrink and turn red even though the MACD line stays above zero: the trend is intact but has lost speed. Read the bars as acceleration and the line as speed, and the pane stops being confusing.

A useful mental picture: you are a passenger in a car. The speedometer is the MACD line. The feeling of being pressed back into your seat is the histogram. You can be travelling at 70 miles an hour (high MACD) with no feeling of acceleration at all (histogram near zero). You can be travelling slowly but feel a strong push (low MACD, growing histogram) as the driver puts their foot down. Neither reading tells you where the car will be in ten minutes.

COMMON MISTAKE

Reading red histogram bars as "bearish". Red bars above a positive MACD line mean an uptrend is slowing down. Slowing is not reversing — most healthy uptrends slow down several times before they end.

What MACD cannot tell you

An honest instrument has an honest specification. MACD measures the relationship between two averages of past closing prices. That means there are things it simply cannot see.

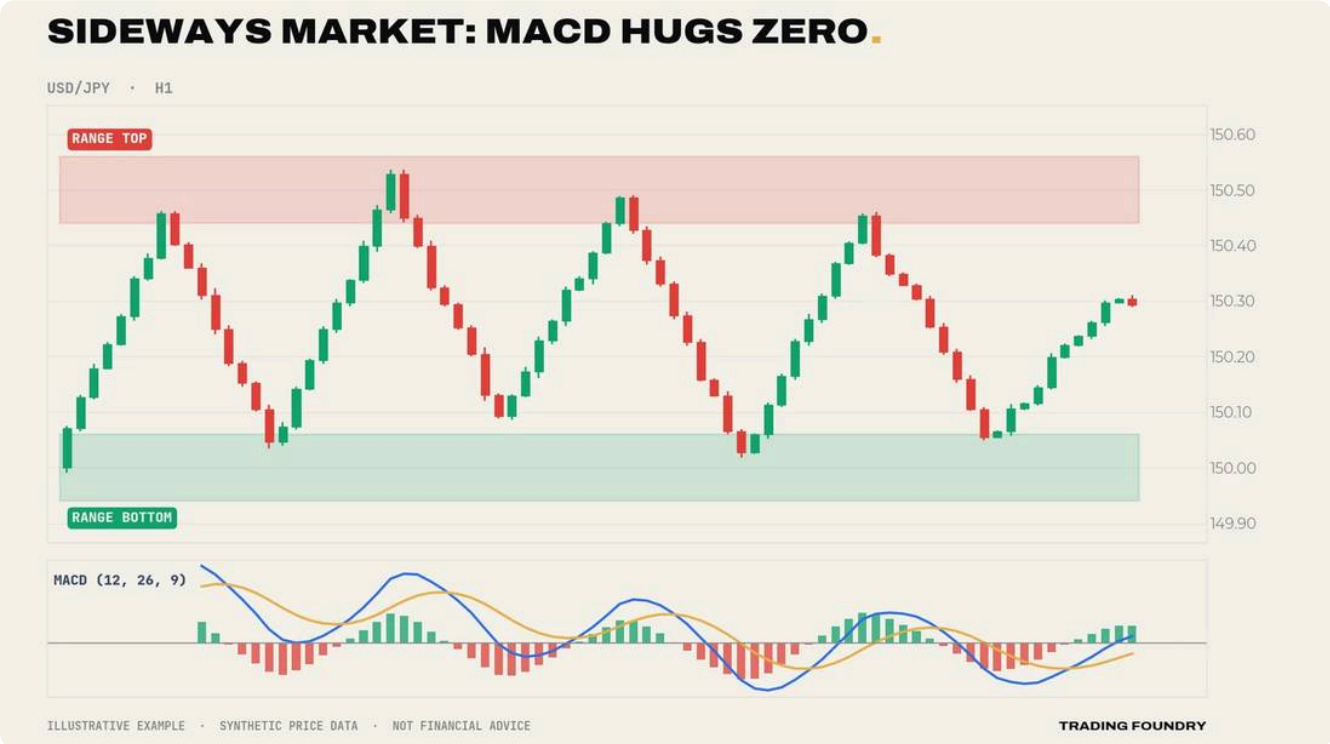
It cannot see levels. MACD has no idea that price is sitting on a major support zone, a prior swing high or a round number. It will happily show strong upward momentum straight into resistance. Levels come from the

price chart.

It cannot see the future. Every input is a past close. When MACD turns, it is because price has *already* turned. That is not a flaw to be fixed with clever settings; it is the definition of an average. Lesson 2 measures exactly how much this lag costs.

It does not know about ranges. In a sideways market, the two averages wind around each other and the MACD line wanders back and forth across zero, producing a stream of crossovers that mean almost nothing. Lesson 8 is devoted to this.

Its values are not comparable across markets. Because the MACD line is measured in price units, a reading of +20 on EUR/USD hourly and +20 on USD/JPY daily mean completely different things. The numbers on the axis matter far less than the shape: rising or falling, above or below zero, growing or shrinking bars.



With no trend, the 12 and 26 EMAs sit almost on top of each other, so the MACD line stays near zero and crosses its signal line every few candles. Each of those crossovers looks like a signal. None of them carries real information about direction. The range itself, drawn on price, is the useful information here.

Reading a MACD pane in four questions

Before any of the strategies later in this course, train your eye with four questions. Ask them, in this order, every time you look at a MACD pane:

1. **Is the MACD line above or below zero?** This tells you whether the 12 EMA is above or below the 26 EMA — a broad description of which side has been in control recently.
2. **Is the MACD line rising or falling?** This tells you whether the trend's speed is increasing or decreasing.

3. **Are the histogram bars growing or shrinking?** This tells you whether that change in speed is itself gaining or losing force.
4. **Does the price chart agree?** Swing highs and lows, support and resistance, and the slope of price always have the final word.

Work through the four charts in this lesson using those questions. You will find you can describe each one in a sentence: "fast uptrend, now slowing", "trend paused then accelerated", "no trend at all". That sentence is what MACD is for. Everything else in this course builds on it.

Going deeper

The histogram is a rate of change of a rate of change. The MACD line approximates the first derivative of a smoothed price — its slope. The signal line is that slope smoothed again over nine candles, which lags it by roughly four candles. So the histogram (MACD minus signal) is approximately four times the recent change in the MACD line: a smoothed second derivative. This is why the histogram turns first, the MACD line turns second, and the zero line is crossed last. The order is mechanical, not a market pattern, and you should expect it every time.

Why EMAs and not SMAs. An exponential average weights recent closes more heavily, so it responds faster and does not "drop" an old price off the end of the window. With simple averages, a big candle leaving the window can move the indicator even when nothing happens today. EMA-based MACD avoids that artefact, which is one reason the default has survived for decades.

Volatility scaling. Because MACD is in price units, a quiet market produces small readings and a volatile market large ones. Some traders divide MACD by price, or by the average true range, to make readings comparable across markets — the "percentage price oscillator" is the first of those. For the discretionary methods in this course you do not need any of that, because every decision is made relative to the same chart's recent history: is the line higher or lower than before, above or below zero, are the bars growing or shrinking?

Start-up distortion. An EMA needs history to settle. The first few dozen candles of any MACD calculation are unreliable, because the averages were seeded on a single price. Platforms handle this by loading plenty of history; it only matters if you are building the indicator yourself or testing on short data sets.

Summary

- The MACD line is the 12 EMA minus the 26 EMA — the gap between a fast and a slow average.
- Read the line as speed: positive and rising means an uptrend that is speeding up; positive and falling means an uptrend that is slowing.
- The signal line is a 9-period average of the MACD line; the histogram is the difference, and reads as acceleration.
- "Signal line crossover" and "histogram changes colour" are the same event.

- MACD cannot see levels, cannot see the future, and produces meaningless crossovers in ranges. Price structure has the final word.

CHECK YOURSELF · ANSWERS AT THE BACK

1. What does the MACD line measure?

   A. The difference between the 12 EMA and the 26 EMA of price

   B. The number of up candles minus down candles

   C. The average of the last 9 closes

   D. Buying volume minus selling volume

2. The MACD line is above zero but falling. Which description is correct?

   A. Price is falling

   B. An uptrend is losing speed

   C. A downtrend is beginning

   D. The histogram must be green

3. A trader says 'the MACD just crossed its signal line to the upside'. What does the histogram show at that moment?

   A. It crosses from below zero to above zero

   B. It reaches its highest bar

   C. Nothing — the histogram is unrelated

   D. It turns from green to red

4. In a steady trend rising 4 pips per candle, roughly where will the MACD line settle with default settings?

   A. About +4

   B. About +28

   C. About +100

   D. It cannot be estimated

5. Price is pressing into a major resistance zone and MACD shows strong upward momentum. What does MACD tell you about the resistance?

   A. It will break, because momentum is strong

   B. It will hold, because MACD is overbought

   C. Nothing — MACD cannot see price levels

   D. It will break only if the histogram is green

Why Crossovers Lag — and What It Costs.

MODULE 1 — WHAT MACD REALLY MEASURES · LESSON 2

Measure the delay built into every MACD crossover, count its cost in pips and stop distance, and recognise the situations where that cost wipes out the trade.

The most popular MACD rule in the world is the signal-line crossover: buy when the MACD line crosses above its signal line, sell when it crosses below. It is simple, it is mechanical, and on a chart of a big trend it looks spectacular. The arrows land near the bottom of every rally and near the top of every decline.

Look closer and they do not land near the bottom. They land *after* it. Every single time. That is not bad luck and it is not a setting you have got wrong. It is built into the arithmetic, and once you understand the arithmetic you can put a number on it — and decide when the price of being late is worth paying.

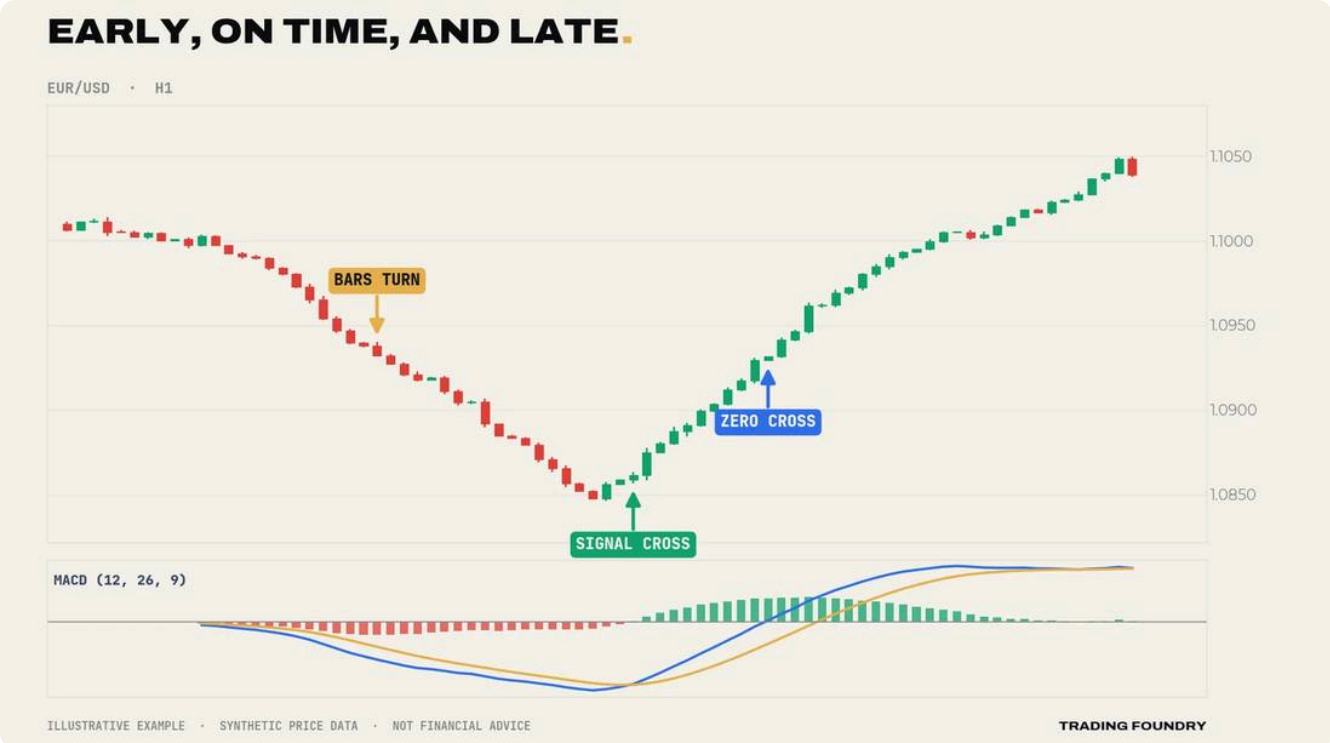
This lesson does that. You will see exactly where the delay comes from, measure it on charts, and learn the three ways it costs you money: a worse entry, a wider stop, and a later exit.

The order of events is fixed

Lesson 1 showed that the MACD line measures speed and the histogram measures acceleration. When a downtrend turns into an uptrend, those readings have to change in a fixed order:

1. **The histogram bottoms first.** The decline stops *accelerating*. Price is still falling, just not faster than before.
2. **The histogram crosses zero.** This is the signal-line crossover. The decline has slowed enough that the MACD line is now rising faster than its own nine-candle average. Price has usually already made its low.
3. **The MACD line crosses zero.** The 12 EMA finally climbs back above the 26 EMA. By now price has typically travelled a good distance off the low.

This sequence is mechanical. It happens in every reversal on every chart. The only thing that changes is how far apart the three events are, and that depends on how the reversal unfolds.



The three MACD events in their fixed order. The histogram stopped falling 17 candles before the low, while price still had about 95 pips to drop — early and wrong for now. The signal-line cross came two candles after the low, 16 pips above it. The MACD line crossed zero 12 candles after the low, about 86 pips up. Early is cheap but unreliable; late is reliable but expensive.

Notice the trade-off in that chart, because it runs through the whole of this course. The earliest event, the histogram turning, was spectacularly early: price kept falling for another 17 candles. Anyone who bought that "momentum shift" sat through nearly 100 pips of loss. The latest event, the zero-line cross, was reliable — by then the trend had clearly changed — but it gave away most of the first leg of the rally. The signal-line cross sat in between.

There is no setting that removes this trade-off. Faster settings move all three events earlier and make each one less reliable. Slower settings do the reverse. Lesson 3 returns to this.

KEY POINT

MACD events always arrive in the same order — histogram, signal cross, zero cross — and each later event is more reliable and more expensive.

Where the lag comes from

The MACD line is built from two exponential averages. An EMA of period N lags a steady trend by roughly $(N - 1) \div 2$ candles: about 5.5 candles for the 12 EMA and 12.5 for the 26 EMA. The signal line is then a 9-period average of that difference, adding roughly another four candles of delay to anything measured against it.

In practice that means a signal-line crossover cannot happen until price has moved far enough, for long enough, to drag the 12 EMA's slope above its recent average. On a sharp V-shaped low, that happens quickly

— one to three candles. On a slow, rounded low, where price drifts sideways for a while before rising, it can take ten candles or more, because nothing in the averages changes until price actually starts moving.

That is the first counter-intuitive fact about crossovers: **they are most punctual in the violent reversals you would have spotted anyway, and least punctual in the slow turns where you most want help.**

Cost one: the worse entry

The obvious cost is price. If the crossover arrives 16 pips above the low, you buy 16 pips higher than the low. That sounds small — and on the chart above, it was small. But that is a 60-pip-per-hour V bottom. On the rounded turns that make up most of the market, the gap is routinely larger, and it is paid on every trade.

Cost two: the wider stop

The less obvious cost is risk. Where does the stop go on a crossover buy? The only structural place is below the low that was just made. Every pip the crossover arrives late is a pip added to the distance between your entry and your stop.

Run the numbers with illustrative values. A trader risks 1% of the account per trade. If the stop is 20 pips away, they can trade a position worth 0.05% of the account per pip. If lag pushes the stop to 40 pips, the same 1% risk buys half the position. The trade can go exactly the same way and pay half as much. Lag does not just cost pips; it shrinks size.

Cost three: the later exit

Traders who buy on crossovers usually sell on crossovers, and the exit lags just as badly as the entry. The result on a whole trend is a chain of costs: in late, out late, and — worst of all — shaken out in the middle.

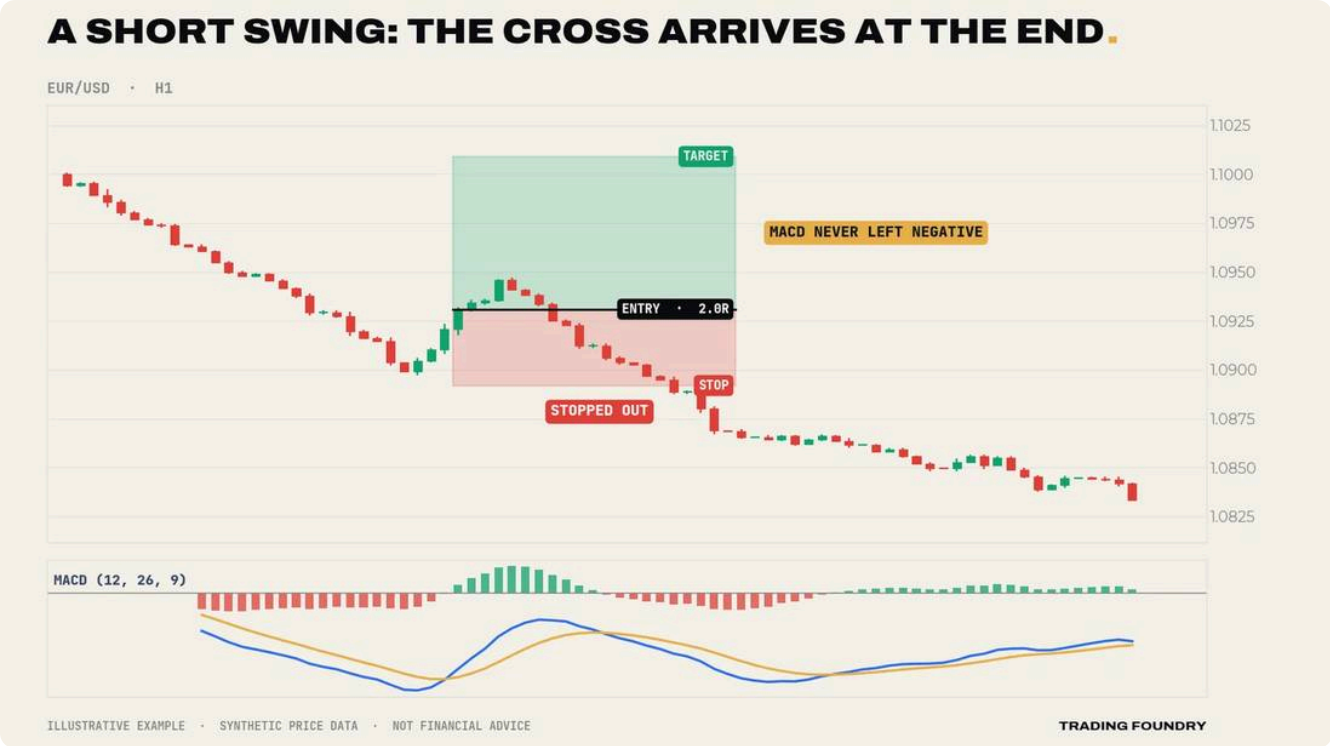


A pure crossover system on one trend. It bought 32 pips off the low, was shaken out by a bearish cross during an ordinary pullback, bought back 32 pips higher than it sold, and exited 25 pips below the top. From a move of about 245 pips it collected about 155, across two trades, paying the spread twice.

The middle of that chart is where crossover systems really hurt. The pullback between the two rallies was an ordinary, healthy correction — the kind Lesson 7 teaches you to *buy*. The crossover system sold it instead, then bought back higher once the trend resumed. Nothing went wrong; the indicator did exactly what its maths dictates. It simply cannot tell the difference between a pullback and a reversal, because both look like "speed falling".

When the lag eats the whole move

On a big trend, late entries and late exits still leave something on the table for you. The dangerous case is the small move. If the swing is only slightly larger than the crossover's built-in delay, the signal arrives right where the move ends.



A bullish signal-line cross arrives 32 pips off a low that is still inside a downtrend, with the MACD line well below zero. The stop must sit under the low, 39 pips away. The bounce is already over; price rolls back down and the stop is hit. Even if the trader had exited on the next bearish cross instead, they would have lost about 25 pips.

This is the losing trade every crossover trader knows. The bounce was real, the crossover was real, and the trade was still a loss, because by the time the signal was confirmed the bounce was nearly finished. Note what the chart was saying all along: the MACD line never got near zero. It was a pause in a downtrend, not a new uptrend. Lesson 4 turns that observation into a filter that would have skipped this trade.

COMMON MISTAKE

Treating a crossover as a timing signal on its own. On small swings the delay is a large fraction of the move, so the signal arrives after most of the opportunity has gone and the stop must be wide.

Practice drill

Use what you have learned so far. The chart below stops at the moment a bullish crossover prints. Decide before you press reveal.

PRACTICE DRILL · BUY, SELL OR WAIT? DECIDE, THEN TURN THE PAGE



The histogram has just turned green and price is pushing back above the prior high. A crossover system says buy. Would you?

THE REVEAL



THE ANSWER

No trade. The cross arrived about 75 pips above the swing low a stop would need to sit under, after a 220-pip run, and the MACD line's new peak is roughly half its first peak — this push is far slower than the first. Late signal, wide stop, fading speed, straight into the prior high. Price stalled and fell back through the swing low.

The answer is in the reveal, but think about how you would have worked it out. Three questions do most of the work: how far is the stop, how much room is left, and is the trend gaining or losing speed? A crossover answers none of them.

Going deeper

Quantifying the delay. You can estimate the lag of a crossover on any chart without software. Count the candles between a clear swing low and the bullish signal-line cross that followed. Do this for the last 20 swings on your chosen market and timeframe and note the average delay and its spread. Then measure the typical swing size in pips. If the average delay costs more than about a third of the typical swing, a stand-alone crossover approach is structurally weak on that market — before spreads.

The rounded-turn problem. An EMA only changes slope when new closes differ from its current value. In a slow, flat bottom, closes hover around the averages, so nothing in the MACD changes until the breakout candle arrives. On these turns the crossover often prints on the same candle as a simple break of the range high — and the range high was the better reference, because it also tells you where the stop belongs.

Why exits hurt more than entries. Entries are optional; you can skip a late one. A crossover exit rule is compulsory, and it fires on every pullback big enough to slow the trend. In strongly trending markets a large

share of the profit comes from staying in through those pullbacks. Many experienced traders therefore use MACD for context and entry timing but manage exits with structure — a close beyond the last higher low, for example — rather than with the opposite crossover.

Expectancy, illustratively. Imagine two hypothetical systems with identical direction calls. System A enters 10 pips after the turn, System B 30 pips after. Average stop distance rises from 25 to 45 pips. If both risk 1% per trade, System B trades about 44% smaller on every position, and its targets are 20 pips closer to wherever the move eventually ends. The numbers are invented; the shape of the problem is not. Lateness compounds through position size.

Summary

- MACD events arrive in a fixed order: histogram turns, signal line crosses, zero line crosses — each later event more reliable and more expensive.
- Lag comes from the averages themselves; no setting removes it, only trades it for noise.
- Lag costs you three ways: a worse entry, a wider stop (and so a smaller position at fixed risk), and a later exit.
- Crossover exits shake you out of healthy pullbacks and force you to re-enter higher.
- On small swings the delay can consume the whole move. A crossover alone is not a strategy.

CHECK YOURSELF · ANSWERS AT THE BACK

1. In a reversal from downtrend to uptrend, which MACD event happens first?
   A. MACD line crosses zero
   B. Signal-line crossover
   C. Histogram stops falling and turns up
   D. They happen at the same time
2. Why does a late crossover entry usually mean a smaller position for the same 1% account risk?
   A. Brokers reduce leverage after crossovers
   B. The stop under the swing low is further away, so fewer units fit inside the same risk
   C. MACD signals require smaller lots by convention
   D. It doesn't — size is unaffected
3. On which type of turn does a signal-line cross typically arrive most promptly?
   A. A slow, rounded bottom
   B. A sharp V-shaped reversal
   C. A sideways range
   D. A gap at the weekly open
4. In the 'In late, out early, back in higher' chart, what caused the mid-trend loss of position?
   A. A genuine reversal
   B. A bearish cross during an ordinary pullback
   C. A spike on news
   D. The zero line was crossed
5. A bullish cross prints 40 pips above the swing low on a market whose typical swing is 60 pips. What is the main problem?
   A. Nothing — a cross is a cross
   B. The delay has consumed most of the typical swing, leaving poor reward for a wide stop
   C. The histogram is the wrong colour
   D. The MACD settings are too slow for that pair
6. Which of these would a crossover-only system be unable to tell apart?
   A. A pullback and a reversal
   B. An H1 chart and an H4 chart
   C. Two different currency pairs

D. A green and a red candle

Settings, Timeframes and the Myths.

MODULE 1 — WHAT MACD REALLY MEASURES · LESSON 3

Choose MACD settings for sound reasons, use timeframe rather than parameters to change what MACD sees, and spot the settings myths that cost traders months.

Sooner or later every MACD user goes looking for better settings. Someone on a forum swears by 5, 35, 5. A course sells "the institutional MACD" with numbers that turn out to be 8, 17, 9. A backtest shows that 10, 23, 7 would have made a fortune on EUR/USD last year. The default 12, 26, 9 starts to look old-fashioned.

This lesson will save you that search. Changing the settings does not change what MACD is. It only moves you along one line — earlier and noisier at one end, later and calmer at the other — and you already have better ways to move along that line. By the end you will know why the defaults are fine, what actually changes when you change them, and why the timeframe you choose matters far more than the numbers you type in.

What the three numbers do

Recall the definition from Lesson 1. In MACD (12, 26, 9):

- **12** is the period of the fast EMA.
- **26** is the period of the slow EMA.
- **9** is the period of the signal line, an EMA of the MACD line.

Shorten the first two and the averages hug price more tightly. The MACD line responds sooner to changes in speed, and it also responds to smaller wiggles that do not matter. Lengthen them and the line becomes smoother, slower and quieter. Shorten the third and the signal line chases the MACD line more closely, producing more crossovers. Lengthen it and crossovers become rarer and later.

That is the entire menu. Every setting you will ever see online is a point on the same trade-off between **responsiveness** and **reliability**. There is no combination that is both earlier and cleaner, because both properties come from the same averaging.

KEY POINT

Every MACD setting is a trade between lag and noise. No setting removes lag without adding noise.

You already have three speeds built in

Before reaching for new numbers, notice that the standard MACD already offers three different speeds. Lesson 2 showed them as a fixed sequence: histogram first, signal-line cross second, zero-line cross last. Choosing which of those events you act on changes the indicator's speed far more than adjusting a parameter from 12 to 10.



The same default MACD read at two speeds. Every pullback in this steady uptrend produced a bearish signal-line cross followed by a bullish one: eight crossovers in total. The MACD line itself never fell below zero. If the signal line is too jumpy for your purpose, read the zero line instead — no new settings required.

That chart is the whole settings debate in one picture. A trader who found the signal-line crosses too noisy might go looking for slower settings. But the zero line on the default indicator already gives a slower, calmer reading of the same trend. Lessons 4 and 5 build methods around exactly this: the zero line for direction, the histogram for timing.

The timeframe is the real setting

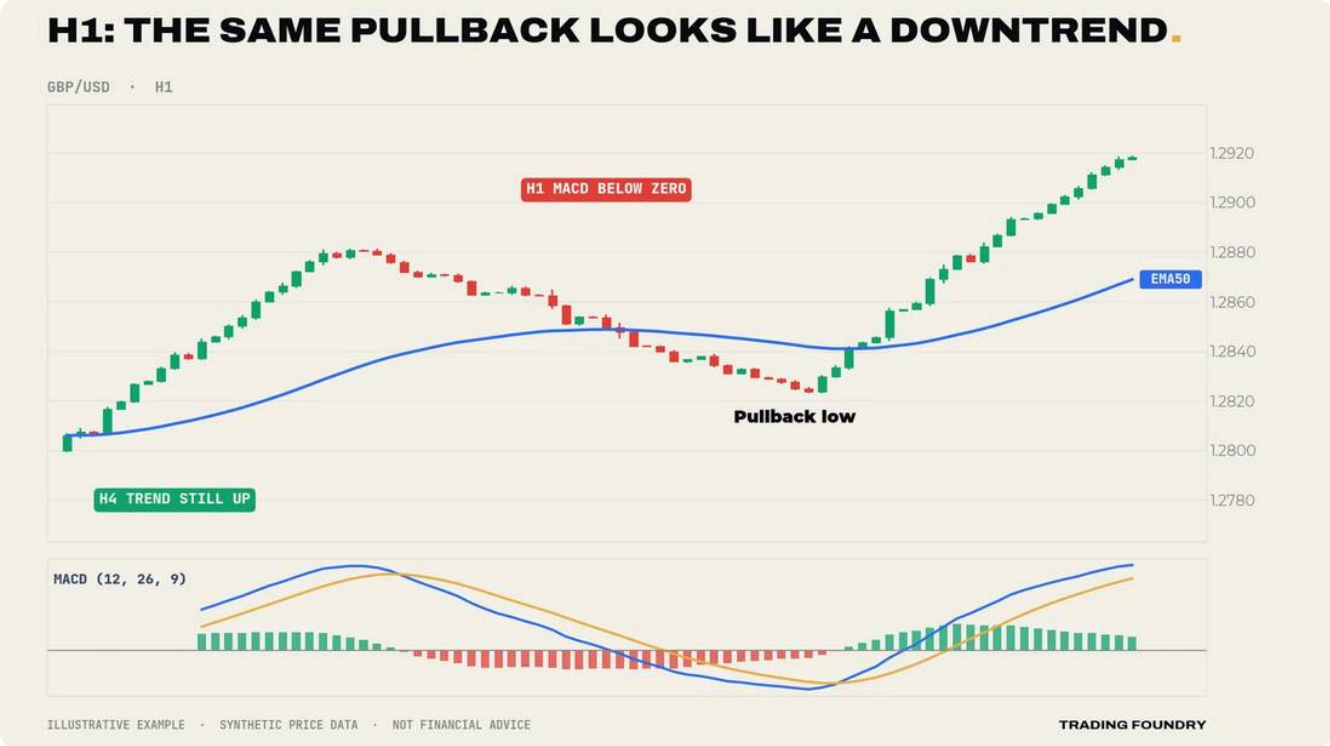
MACD does not know what a candle is. It counts them. A 12-period EMA on an hourly chart averages the last 12 hours; on a four-hour chart it averages the last 48 hours. So changing the timeframe changes what the indicator measures far more dramatically than changing the parameters.

There is a useful rule of thumb here. Each four-hour candle contains four hourly candles, so MACD (12, 26, 9) on the H4 chart sees roughly what MACD (48, 104, 36) would see on the H1 chart. If you find yourself wanting much slower settings on a lower timeframe, what you probably want is the higher timeframe, viewed with its normal settings. You will get a cleaner picture, with candles that carry their own information, rather than a stretched-out average of noise.



On the four-hour chart this market is in a clean uptrend. The pullback circled near the right is a minor dip; the MACD line eases but stays comfortably above zero. At this timeframe, there is nothing to worry about.

Now look at the same kind of pullback through an hourly lens.



Zoomed in to hourly candles, the pullback lasts long enough to drag the H1 MACD line below zero for roughly twenty candles. Read alone, the hourly chart says 'downtrend'. Read with the four-hour chart, it says 'pullback in an uptrend' — which is what it turned out to be.

Neither chart is wrong. They answer different questions. The higher timeframe tells you which direction you want to trade; the lower timeframe tells you whether the pullback within it is still running or has started to turn. This is the basis of the multi-timeframe approach used later in the course: **direction from the higher timeframe, timing from the lower one**, and the standard settings on both.

COMMON MISTAKE

Changing MACD settings to stop the lower-timeframe indicator from "disagreeing" with the higher timeframe. The disagreement is the information. A pullback on the higher timeframe is supposed to look like a small downtrend on the lower one.

Five myths, briefly

Myth 1: there is an optimal setting. You can always find settings that would have performed best on a past data set. That tells you about the past data set, not about next month. The more parameters you optimise, and the more finely you tune them, the more you are fitting the noise of that sample. Settings that only work at exactly 11, 24, 8 and fall apart at 12, 26, 9 are not an edge; they are an accident.

Myth 2: faster settings fix the lag. They move every signal earlier — including the false ones. On the chart above, a faster MACD would add crossovers during every small wiggle inside each pullback. You exchange a late signal for an early signal that is wrong more often.

Myth 3: MACD has overbought and oversold levels. It does not. Unlike RSI, which is boxed between 0 and 100, MACD is unbounded and measured in price units. A "high" reading is only high compared with the same market's recent readings. There is no universal level at which MACD is "too high".

Myth 4: some currency pairs need special settings. Volatility differs between pairs, so the size of the readings differs. The shape of the readings — rising, falling, above or below zero — means the same thing everywhere. Read shape, not size.

Myth 5: the default settings are arbitrary. They are conventional, which is a different thing. Because almost every platform loads 12, 26, 9, almost every trader looking at MACD is looking at the same picture. That does not make it magic, but it does mean you are reading the same instrument as everyone else, and your observations, notes and backtests transfer between platforms without translation.

MACD reacts to size, not significance

One last point about what the arithmetic does, because it explains a lot of strange MACD behaviour. Because the indicator is built from averages of closes, one enormous candle has an enormous effect on it. A single spike on a data release can produce the largest MACD reading on the chart, even if price gives the whole move back within a day.



An 80-pip candle on a data release produced the strongest MACD reading on this chart and a crisp bullish signal. Within fifteen candles about 80% of the move had been retraced, and the rest soon followed. MACD measured the size of the candle accurately; it has no way to measure whether that move had any follow-through behind it.

The practical rule: treat MACD readings formed by one or two outsized candles with suspicion. Wait for the indicator to reflect a run of candles rather than a single event, and let the price chart — does the breakout hold? does the next pullback stay above the spike's base? — decide whether the move was real.

Going deeper

Lag, quantified. An EMA with period N has a smoothing constant of $2 \div (N + 1)$ and an average lag of about $(N - 1) \div 2$ bars on a steady trend. Halving the fast and slow periods roughly halves the lag of the MACD line — and roughly doubles its sensitivity to single-bar noise. If you test alternative settings, test whole families (for example 8/17/9, 12/26/9 and 19/39/9) and look for broadly similar behaviour across them. Robust ideas survive nearby settings; fragile ones do not.

Choosing a timeframe pair. A ratio of about four to six between timeframes works well: D1 with H4, H4 with H1, H1 with M15. Closer than that and the two charts show nearly the same thing; further apart and the lower timeframe's swings are too small to matter to the higher one.

When a non-default setting is defensible. If you trade a market with very different session structure — one that is only active for a few hours a day, for example — the number of candles per meaningful swing may genuinely differ. The right test is to count candles in typical swings, not to optimise for profit. If the typical swing on your chart lasts 30 candles, a slow EMA of 26 sits sensibly inside it. If typical swings last 8 candles, consider a lower timeframe rather than shorter settings, so each swing contains more candles.

Testing honestly. If you must test settings, split your data. Choose settings on the first half, then run them unchanged on the second half you have not looked at. If performance collapses out of sample, the settings were fitted to noise. Write the result down either way; most traders only remember the tests that worked.

Summary

- The three numbers set the fast EMA, slow EMA and signal line. Every combination trades lag against noise.
- The default MACD already offers three speeds: histogram, signal-line cross and zero-line cross. Choose the event before you change the numbers.
- Timeframe changes MACD far more than settings. H4 with default settings is roughly H1 with settings four times longer — and cleaner.
- A lower-timeframe MACD disagreeing with a higher-timeframe one is information, not a fault.
- MACD has no overbought or oversold levels, and single large candles can dominate it.

CHECK YOURSELF · ANSWERS AT THE BACK

1. What happens if you shorten the fast and slow periods of MACD?

   A. Signals arrive earlier and the indicator becomes noisier

   B. Signals arrive earlier and become more reliable

   C. Lag disappears entirely

   D. The indicator gains overbought and oversold levels

2. Default MACD on H4 sees roughly the same history as which setting on H1?

   A. MACD (3, 6, 2)

   B. MACD (12, 26, 9)

   C. MACD (48, 104, 36)

   D. MACD (24, 52, 18)

3. In the 'One trend, eight crosses' chart, which reading gave the calmest picture of the trend?

   A. The histogram

   B. The signal-line crossovers

   C. The MACD line relative to zero

   D. None of them

4. The H1 MACD is below zero while the H4 MACD is comfortably above zero. What is the most sensible reading?

   A. The indicator is broken on one timeframe

   B. A pullback within a higher-timeframe uptrend

   C. A confirmed downtrend on both timeframes

   D. Change the H1 settings until they agree

5. A MACD reading of +0.0040 on EUR/USD — is the market overbought?

   A. Yes, above +0.0030 is overbought

   B. No, MACD has no fixed overbought level; it is unbounded and in price units

   C. Yes, if the histogram is green

   D. Only on the daily chart

6. A single 80-pip news candle produces the largest MACD reading on the chart. How should you treat it?

   A. As the strongest buy signal on the chart

   B. With suspicion until price shows follow-through

   C. As proof of institutional buying

D. As a reason to change settings

The Zero Line as a Trend Filter.

MODULE 2 — READING MACD PROPERLY · LESSON 4

Use the MACD zero line to decide which direction you are allowed to trade, ignore every crossover that fights it, and know the two situations where the filter itself goes wrong.

Lesson 2 ended with a losing trade: a bullish crossover that fired in the middle of a downtrend, bought a bounce that was already ending and was stopped out. The chart had been telling the truth all along. The MACD line never got near zero. The bounce was a pause, not a new trend.

That observation is the most useful thing you can do with MACD, and almost nobody uses it properly. The zero line is not a signal. It is a **filter** — a way of deciding, before you look for any entry at all, which side of the market you are allowed to be on. This lesson shows you how to use it, how much bad trading it removes, and where its own lag will still catch you out.

What the zero line really says

The MACD line is the 12 EMA minus the 26 EMA. So the MACD line is above zero exactly when the 12 EMA is above the 26 EMA, and below zero exactly when it is below. That gives the zero line a simple meaning:

- **Above zero:** over the last few weeks of candles, recent prices have been higher than older prices. Buyers have had the better of it.
- **Below zero:** recent prices have been lower than older prices. Sellers have had the better of it.

That is a crude definition of trend, but crude is what you want from a filter. It does not need to be clever. It needs to be consistent, and it needs to stop you doing the most expensive thing a short-term trader does: fighting the prevailing direction because a small counter-move looked convincing.



For the first two thirds of the chart the MACD line sits below zero: the filter allows short trades only. The circled candle is where the 12 EMA finally climbs back above the 26 EMA and the MACD line crosses zero. From there the filter allows longs only. Notice how late that switch is — almost twenty candles after the low.

The rule

The zero-line filter is one sentence:

KEY POINT

Trade longs only while the MACD line is above zero. Trade shorts only while it is below zero. Everything else is ignored.

Notice what this rule does *not* say. It does not say "buy when MACD crosses above zero". The zero cross is not an entry; it is the moment the permission changes. Entries come from elsewhere — the histogram, a pullback to a moving average, a price trigger — and the rest of this course teaches them. The filter's only job is to remove half of them before you even consider them.

What the filter throws away

Here is the same downtrend kind of market that produced Lesson 2's losing trade. Count the bullish crossovers.



Four bullish signal-line crosses (circled), every one of them below zero, every one of them a short-lived bounce in a downtrend. The zero-line filter ignores all four. The bearish crosses that follow each bounce agree with the filter and are allowed — although, as you will see in Lesson 7, there are better places to sell than the crossover itself.

Four signals that a crossover trader would have bought are simply not available to a filtered trader. Each of those bounces looked like the start of something at the time. None of them was. That is the filter's value: it does not make you better at reading bounces, it stops you having to.

Now look at the allowed side. The bearish crossovers fired near the top of each bounce, with the MACD line below zero. They still carry all of the lag problems from Lesson 2, but they are pointing in the same direction as the trend, so the trend does some of the work of rescuing a late entry.

A filtered trade, start to finish

The filter becomes useful when you combine it with a sensible entry and a structural stop. Here is the simplest version: a with-trend crossover, taken only because the filter allows it.



An uptrend pulls back. The MACD line dips but stays above zero, so the filter keeps longs available. When the histogram turns green again, the long is taken with the stop below the pullback's higher low and a target at twice the risk. The trend resumes and the target is reached.

Walk through the numbers, which are illustrative. The bullish cross confirmed at 133 pips on the chart's scale. The higher low it formed above sits at 83, so the stop goes at 78, just beneath it: 55 pips of risk. A target at 2R is 110 pips above entry. Risking 1% of the account, a 55-pip stop sets the position size; the target, if reached, returns 2%. The trade worked, but notice the entry was still 50 pips above the low. The filter fixed the direction, not the lag.

When the filter is wrong

A filter built from moving averages has the same weakness as the averages: it is late. There are two situations where that lateness turns into losing trades, and you need to recognise both.

The first is the regime change. After a long trend, the MACD line can stay on the "allowed" side of zero for some time after price has already turned. During that window, the filter keeps permitting trades in a direction that structure has already abandoned.



A bullish cross with the MACD line still above zero: the filter says longs are allowed. But price has just made a lower high after a deep, fast drop, and the MACD line's peak is a third of the first one. The trade is stopped out as the zero line is finally crossed. The filter lagged the change in structure.

That trade obeyed the rule and still lost. What would have saved it? Price structure. The decline from the high was fast and deep — over 40% of the whole rally in ten candles — and the bounce failed below the high. Any trend-following trader would call that a warning, and Lesson 9 builds it into the rules. MACD's filter is a minimum condition, never a sufficient one.

The second is the range. When a market goes sideways, the 12 and 26 EMAs wind around each other and the MACD line crosses zero again and again. The filter flips from "longs only" to "shorts only" every few candles, which is another way of saying it has stopped working. Lesson 8 shows how to recognise this early and stand aside.

COMMON MISTAKE

Treating the zero cross itself as an entry. By the time the 12 EMA crosses the 26 EMA, a new trend is usually well under way and the first pullback has not happened yet. Wait for the pullback.

What to do when the filter switches

The moment the MACD line crosses zero is a change of permission, so treat it as an administrative event rather than a trading one. Three things happen. First, any open position on the old side is reviewed: if its own stop or target has not been reached, the trend reason for holding it has gone, and Lesson 9 makes exiting it a rule. Second, you stop looking for setups on the old side. Third, you start waiting — not trading — on the new side. A fresh trend needs to show you an impulse, a pullback and a reset before the methods in Module 3 have

anything to work with. In practice that wait is often ten to thirty candles. It feels like missing the move. It is actually the price of trading only the parts of a move you can define.

Going deeper

Zero line versus a slower moving average. The zero-line filter is equivalent to asking whether the 12 EMA is above the 26 EMA. Many traders use price relative to a 50 or 200 EMA as their trend filter instead. These are close cousins. The MACD version reacts sooner than a 200 EMA filter and is noisier; it sits roughly alongside a 50-period average in speed. If you already use a moving-average filter, adding the MACD zero line on the same timeframe will mostly agree with it and add little. The genuinely useful addition is the **higher-timeframe** MACD zero line, which Lesson 3 introduced.

A two-timeframe filter. A robust arrangement is: higher-timeframe MACD line above zero sets the direction; trading-timeframe MACD is then used for timing pullbacks within that direction. The trading-timeframe MACD will fall back towards zero during pullbacks, and sometimes slightly below it — Lesson 7 builds its setup around that fall towards zero. What is not allowed is a long while the higher-timeframe MACD is below zero.

Distance from zero. The further the MACD line is from zero, the faster the trend has been. A very large reading after a long run can mean the move is stretched, but MACD has no fixed extreme, so judge this only against the same chart's history. A practical habit: when the MACD line is at its most extreme reading for the last hundred candles, avoid opening new with-trend positions until there has been a pullback that brings it back towards its average.

Measuring what the filter removes. If you keep a trading journal, tag every crossover signal for a month as "with filter" or "against filter" without trading them. Then compare the two groups on a simple measure — how far price moved in the signal's favour before it moved 20 pips against it, for example. Your own data will tell you more about the filter than any book can. Expect the difference to be largest in trending months and smallest in ranging months.

Summary

- MACD above zero means the 12 EMA is above the 26 EMA; below zero means the reverse. It is a crude but consistent definition of trend.
- The filter: longs only above zero, shorts only below. Ignore every crossover that fights it.
- The zero cross is a change of permission, not an entry.
- The filter removes most counter-trend bounce trades but does not cure lag.
- It fails at regime changes (it stays open after structure breaks) and in ranges (it flips constantly). Structure overrides it.

CHECK YOURSELF · ANSWERS AT THE BACK

1. The MACD line is below zero and a bullish signal-line crossover prints. What does the zero-line filter say?

   A. Buy — crossovers override the filter

   B. Ignore it — only shorts are allowed below zero

   C. Buy half size

   D. Close all short positions

2. What does it mean, in terms of moving averages, when the MACD line is above zero?

   A. Price is above the 200 EMA

   B. The 12 EMA is above the 26 EMA

   C. The histogram is green

   D. RSI is above 50

3. Why is the MACD zero-line cross a poor entry signal on its own?

   A. It never happens in trends

   B. It arrives late, typically well after the turn and before the first pullback

   C. It is only valid on the daily chart

   D. It repaints

4. In 'The filter is still open — but the trend is not', what warned against the long despite the filter allowing it?

   A. The histogram was green

   B. A deep, fast drop and a lower high, with a much smaller MACD peak

   C. The pair was USD/JPY

   D. The MACD line was above zero

5. A trader's filter flips between 'longs only' and 'shorts only' six times in forty candles. What is the most likely market condition?

   A. A strong uptrend

   B. A strong downtrend

   C. A sideways range

   D. A news spike

Reading Histogram Momentum Shifts.

MODULE 2 — READING MACD PROPERLY · LESSON 5

Use the MACD histogram to see a trend's pushes weakening, time the end of a pullback with the histogram hook, and avoid the false alarms that red bars produce in healthy trends.

The histogram is the part of MACD most traders glance at and least understand. It changes colour often, it moves first, and it looks like it ought to be telling you something urgent. Usually what it is telling you is far more modest than it looks — and occasionally it is the earliest honest warning on the whole chart.

Lesson 1 described the histogram as acceleration: the MACD line's speed reading compared with its own nine-candle average. This lesson turns that description into three practical skills. You will learn to compare the strength of successive pushes, to use the **histogram hook** to time the end of a pullback, and to stop treating every red bar in an uptrend as an alarm.

Bars are about change, not direction

Start with the single most important fact about the histogram. It does not tell you which way price is going. It tells you whether the MACD line — speed — is rising or falling relative to its recent average.

That means a histogram can be red while price rises, and green while price falls:

- **Red bars in an uptrend** mean the uptrend's speed is falling below its recent average. Price may still be rising, just more slowly than it was.
- **Green bars in a downtrend** mean the decline is slowing. Price may still be falling.

Once you accept this, a lot of apparently contradictory MACD behaviour stops being confusing.



The trend opens with a burst and then settles into a steady, slower climb. Because the climb is slower than the opening burst, the MACD line drifts lower and the histogram stays red for 37 candles — while price rises roughly another 120 pips. Nothing reversed. The trend simply stopped accelerating.

Anyone who sold that chart because "the histogram turned bearish" sold a trend that had more than a hundred pips left to run. The histogram was accurate: speed was falling. It was the interpretation that was wrong.

KEY POINT

The histogram measures change in speed, not direction. Red bars in an uptrend mean slowing, not falling.

Skill one: comparing pushes

The first useful job for the histogram is comparing the strength of successive pushes in a trend. Each impulse leg in a healthy trend should be accompanied by a cluster of growing bars. If each new push produces smaller bars than the last, the trend is spending more effort for less acceleration.



Three higher highs, each one reached with less acceleration. The green cluster under the first push is tall; under the second it is a fraction of the size; under the third it barely clears zero. Price structure is still bullish, but the trend is spending more effort for less progress — a reason to be selective with new longs, not a reason to sell.

Read this the right way round. Shrinking histogram peaks across successive pushes do not mean the trend is over; price has made three higher highs and no lower low. They mean the trend is **maturing**. The practical consequences are defensive: take profits a little sooner, be pickier about new entries, and pay close attention to the next pullback. If it is deeper and faster than the previous ones, the warning is gaining weight. Lesson 6 develops this same idea into divergence, using the MACD line rather than the histogram.

Skill two: the histogram hook

The histogram's most practical use is timing. During a pullback in a trend, the histogram grows on the counter-trend side as the pullback gathers pace. When the pullback starts to lose force, the bars stop growing and begin to shrink. The first bar that is clearly shorter than the one before it is called the **hook**.

The hook is an early event. It comes before the signal-line cross and well before any zero-line event. That is its value — and, as Lesson 2 warned, its danger. Early signals are cheap but unreliable. So the hook is never an entry on its own. It is a cue to start watching for a price trigger.

The rule for a long is:

1. The trend filter allows longs (MACD line above zero, or on the higher timeframe as in Lesson 4).
2. Price has pulled back and the histogram has gone red.
3. A histogram bar prints that is clearly shorter than the previous red bar — the hook.
4. A price candle closes above the high of the previous candle. That is the trigger. Enter on its close.

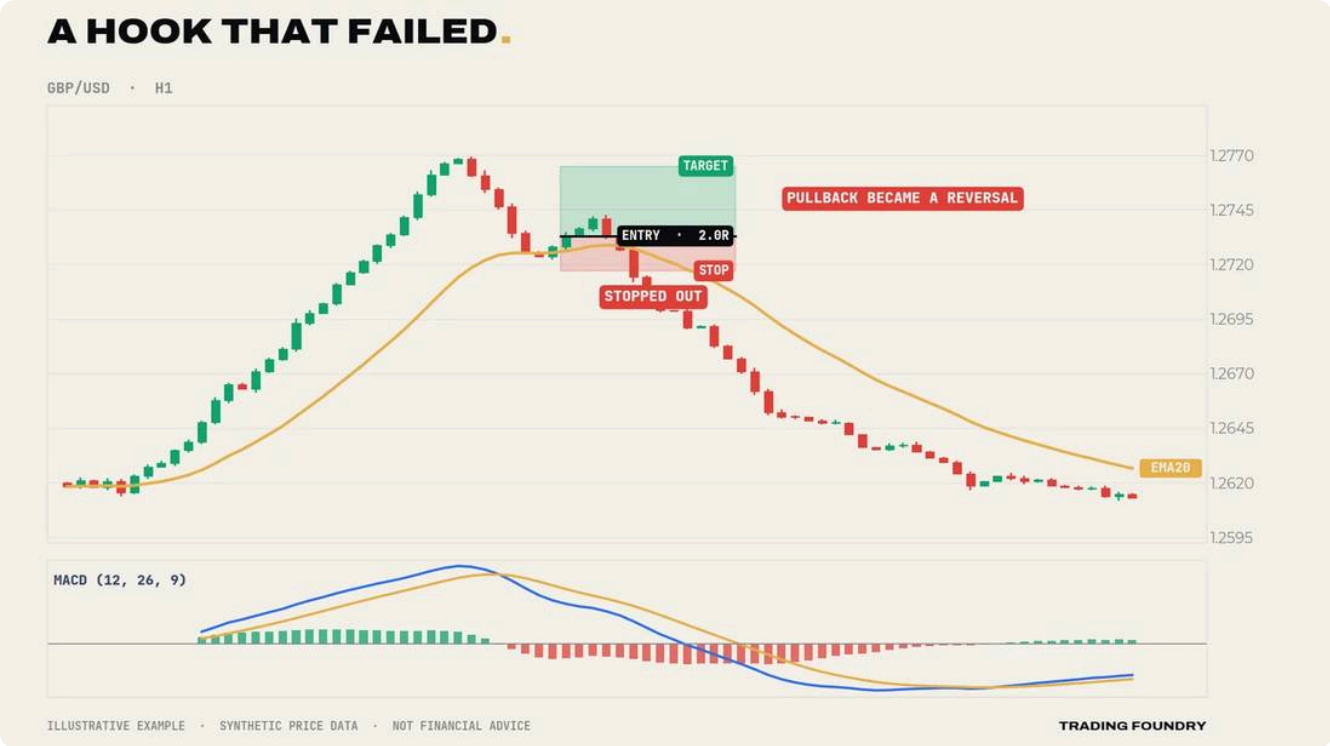
5. Stop goes below the pullback's low. Target at least twice the risk, or the prior swing high if that is closer and still gives at least 1.5R.



The pullback's red bars peak and the first clearly shorter bar prints as price closes back above the previous candle's high. Entry is on that close at 111, stop below the pullback low at 88, target at 2R. The signal-line cross does not arrive until six candles later, 31 pips higher. The hook bought the same trend change considerably cheaper.

The comparison with the crossover is the point. Both the hook trader and the crossover trader were buying the same pullback in the same trend. The hook trader got in 31 pips cheaper and so needed a stop 23 pips wide rather than more than 50. At the same 1% risk, their position was more than twice the size. That is the real reward for acting earlier.

The price is that early signals fail more often. Here is one that did.



The setup looks the same: trend up, MACD above zero, red bars shrinking, a close above the prior candle's high. Price bounces about 10 pips, rolls over and takes out the pullback low. The stop limits the damage to 1R. The warning was in the pullback itself: it fell about 48 pips in six candles — faster per candle than the rally it was correcting — with no pause.

The stop did its job. One loss of 1R against a 2R target is an affordable outcome as long as the method wins often enough over many trades — and you can only find out whether it does by testing it, not by reading about it. What you can do here is look for the clue that would have made you more cautious: a pullback that was steep, fast and unbroken, falling faster per candle than the rally had risen. Pullbacks that look like that deserve a stronger trigger or no trade. Lesson 7 builds that into a full method.

COMMON MISTAKE

Buying the hook without a price trigger. A shrinking red bar only means the decline is slowing. Price can slow and keep falling. Wait for a candle to close above the previous candle's high.

Skill three: ignoring the noise

Most of the time, the histogram is doing nothing worth acting on. In a quiet trend, the bars wobble around zero, flip colour every few candles and mean very little. A useful habit is to ignore the histogram entirely unless one of two things is true: price is in a pullback inside a trend you are allowed to trade (skill two), or you are comparing pushes at a new swing high or low (skill one). Outside those two moments, look at the price chart.

Going deeper

Why the hook comes first. The histogram is MACD minus its 9-period EMA. Because the signal line lags the MACD line by about four candles, the histogram is roughly proportional to how much the MACD line has

changed over the last few candles. When the pullback's speed stops increasing, the MACD line's descent flattens and the histogram contracts immediately. The signal cross needs the MACD line to actually rise past its average, which takes several more candles. The zero line needs the 12 EMA to cross the 26 EMA, which takes longer still.

Defining "clearly shorter". Tiny contractions happen constantly, and some will reverse on the next bar. A practical threshold is that the hook bar must be at least 10–15% shorter than the largest bar of the pullback. Alternatively, require two consecutive shorter bars. Both reduce false hooks; both cost a few pips of entry. Write down which you use and do not switch mid-test.

Pullback speed versus impulse speed. The histogram cannot tell you on its own whether a pullback is gentle or violent, because the histogram's swings are naturally large around any turn. Price can. Divide the impulse's size in pips by the number of candles it took, and do the same for the pullback. On the winning chart above the rally covered about 150 pips in 22 candles, roughly 7 pips per candle, and the pullback about 50 pips in 12 candles, roughly 4 per candle: slower than the trend, as a healthy correction should be. On the failed chart the pullback covered about 48 pips in six candles, 8 per candle: faster than the trend it was correcting. Many traders simply skip hook setups when the pullback is travelling faster than the impulse did.

Lower-timeframe histogram for higher-timeframe pullbacks. If the H4 chart is in a pullback, the H1 histogram will produce several hooks inside it. Using the H1 hook as a trigger only when the H4 pullback has reached a sensible area — a moving average or prior structure — is a way to refine entries without changing the logic. Do not take every H1 hook; take the one that occurs where the H4 chart says a pullback should end.

Summary

- The histogram measures change in the MACD line's speed. Red bars in an uptrend mean slowing, not falling.
- Shrinking histogram peaks across successive pushes mean a maturing trend: be selective and defensive.
- The hook — the first clearly shorter bar of a pullback — is an early cue, not an entry. Wait for a price trigger.
- Hook entries are cheaper than crossover entries, allowing tighter stops and larger size at the same risk, at the cost of more false starts.
- Outside pullbacks and new swing extremes, the histogram is mostly noise.

CHECK YOURSELF · ANSWERS AT THE BACK

1. The histogram has been red for 30 candles while price has risen 150 pips. What is the best reading?
   A. A hidden downtrend
   B. The uptrend continues but at a slower speed than earlier
   C. The indicator is broken
   D. A short signal
2. What is the histogram hook?
   A. The first histogram bar of a pullback that is clearly shorter than the one before it
   B. A histogram bar that crosses zero
   C. The tallest bar on the chart
   D. A hand-drawn line on the histogram
3. Why is a price trigger required after the hook?
   A. Because the hook is late
   B. Because a slowing decline can continue falling; the trigger shows buyers have actually taken control for a candle
   C. Because brokers require it
   D. It isn't required
4. In 'The hook: earlier than the crossover', how much cheaper was the hook entry than the signal-line cross?
   A. About 5 pips
   B. About 31 pips
   C. About 100 pips
   D. It was more expensive
5. Three successive pushes make higher highs with smaller and smaller histogram clusters. What should a trend trader do?
   A. Short immediately
   B. Treat the trend as maturing: be more selective with new longs and protect open profits
   C. Double the next long position
   D. Ignore it — histogram height has no meaning
6. An impulse rose 140 pips in 20 candles. The pullback has fallen 60 pips in 5 candles and the histogram has just hooked. How should you treat the setup?
   A. As a strong buy — the hook is early
   B. With caution or not at all — the pullback is travelling faster per candle than the impulse did

C. As a short signal

D. Wait for the zero-line cross before buying

MACD Divergence Without the Myths.

MODULE 2 — READING MACD PROPERLY · LESSON 6

Identify genuine MACD divergence, understand why it happens, and trade it only after price structure confirms — while recognising the strong trends where divergence repeats and fails.

Divergence is the most romanticised idea in technical analysis. Price makes a new high, the indicator does not, and a story is born: the move is "running out of steam", "smart money is distributing", "a reversal is imminent". Screenshots of divergences that preceded big turns are shared endlessly. Screenshots of divergences that preceded another 300 pips of trend are not.

This lesson treats divergence for what it actually is: a measurement. You already know from Lesson 1 that the MACD line is a speed reading. Divergence is simply what you see when price keeps making progress while that speed reading falls. It is real information, and it is useful — but it is a warning, not a trigger, and in a strong trend it can repeat several times before anything happens.

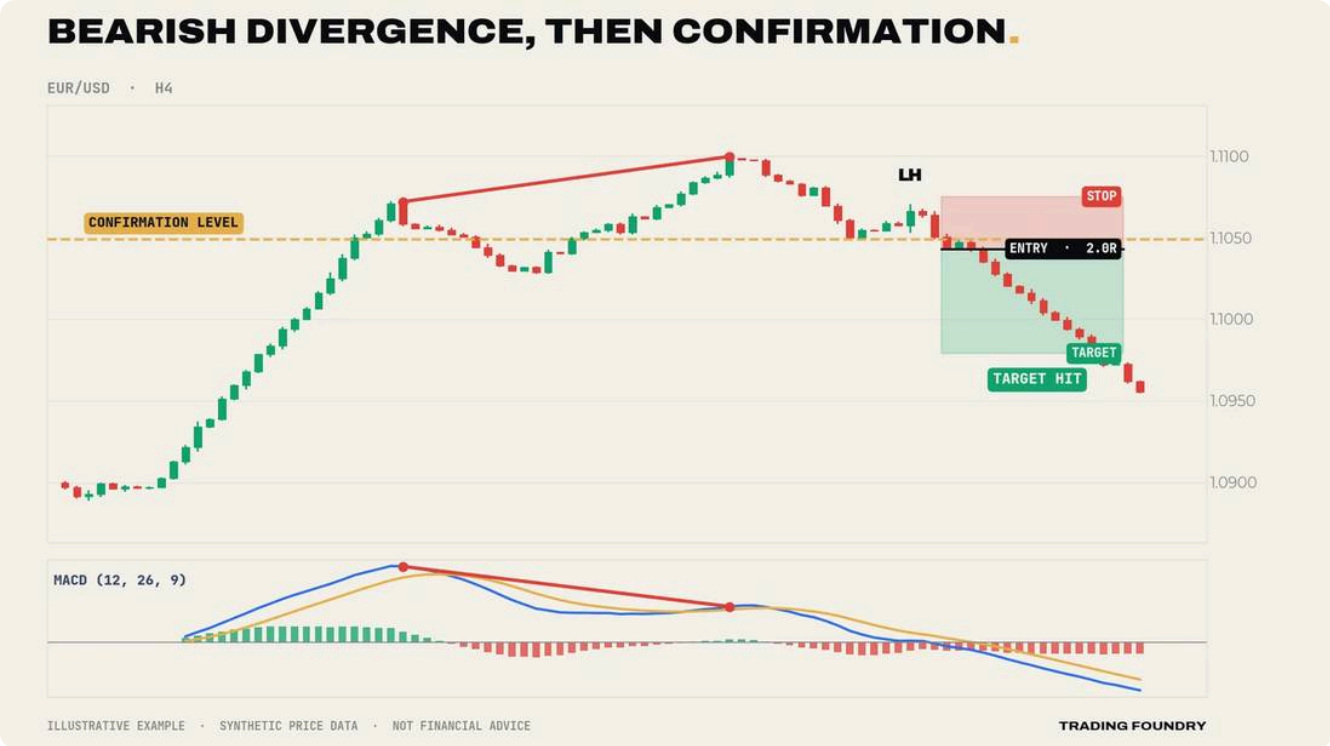
What divergence is

There are two classic forms:

- **Bearish divergence:** price makes a **higher high**, but the MACD line makes a **lower high** at the same time.
- **Bullish divergence:** price makes a **lower low**, but the MACD line makes a **higher low**.

Use the MACD line, not the histogram, for this. The histogram is noisier and produces divergences constantly. The line is smoother and its peaks correspond more closely to the swings you can see on price.

In speed terms, bearish divergence says: "The second push to a new high was slower than the first." Nothing more. The market is still going up — it made a new high — but it got there with less urgency. That might be the first sign of a trend ending. It might also be a trend settling from a sprint into a jog, as the 'Speed peaks before price does' chart in Lesson 1 showed.



Price makes a higher high while the MACD line makes a clearly lower high: bearish divergence. Nothing is done yet. Price then drops, bounces to a lower high (LH) and closes below the low between them — the confirmation level. The short is taken on that close, with the stop above the lower high and a 2R target.

Look carefully at the sequence in that chart. The divergence was visible at the second peak, but the trade did not happen there. It happened about eighteen candles later, after price itself had done two things: made a lower high, and broken the swing low beneath it. By then the divergence was no longer a theory about slowing momentum. It was a slowing trend that had also changed structure.

KEY POINT

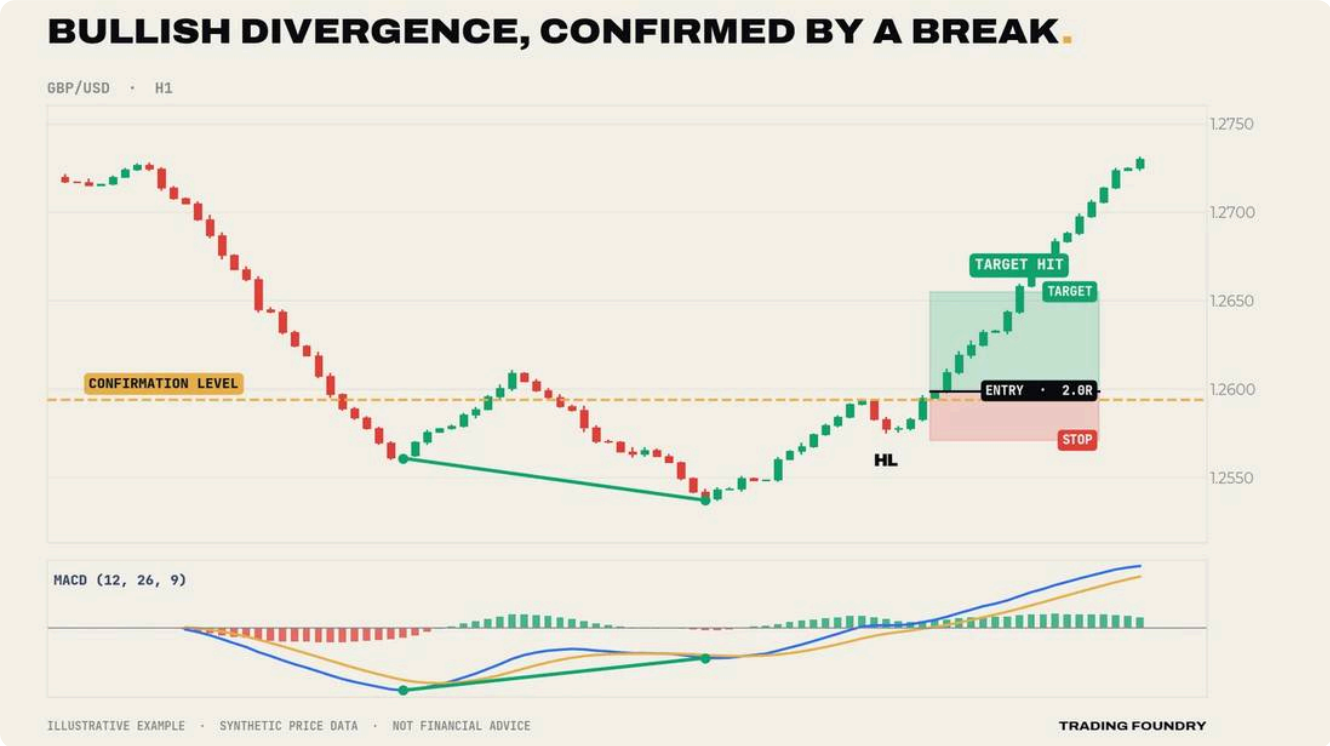
Divergence is a warning, not an entry. Price structure — a lower high and a break of the low beneath it — is the confirmation.

The confirmation rule

For a bearish divergence trade, the full sequence is:

1. **Divergence:** price makes a higher high; the MACD line makes a lower high. Draw the line between the two peaks on both price and indicator.
2. **First drop:** price falls from the second peak and forms a swing low.
3. **Lower high:** price bounces but fails below the second peak.
4. **Trigger:** a candle closes below the swing low from step 2. Enter on that close.
5. **Stop:** above the lower high from step 3.
6. **Target:** at least 2R, or the next significant support if it offers at least 1.5R.

Mirror every step for a bullish divergence. Some traders will think this is too slow, and they are right that it gives away part of the move. What it buys in return is that every trade you take has already shown the two things a reversal needs: lost momentum *and* changed structure. You also get a logical place for the stop, which a raw divergence entry does not give you.



A lower low on price, a higher low on the MACD line. After the second low, price bounces, pulls back to a higher low (HL), then closes above the bounce high. Entry on that close at 99, stop below the higher low, target at 2R. The divergence warned; the break confirmed.

The bullish example has the same shape upside down. Note the stop distance: 28 pips, because the higher low gave a nearby, logical place for it. Compare that with buying at the second low on divergence alone, where the only possible stop is "somewhere below the low" — and where, if the low breaks, you have no idea how far it might run.

When divergence fails

Here is the chart that the divergence screenshots never show. Strong trends decelerate in stages, and every stage can produce a divergence.



Each new high comes with a lower MACD peak — two bearish divergences in a row. A trader who sold the second peak on divergence alone was stopped out within a dozen candles, and the trend went on to add more than 100 pips. The confirmation rule never triggered: price never made a lower high and never broke a swing low.

This is why the confirmation rule matters. The trader who sold on divergence alone was wrong twice — and in a real market, could have been wrong three or four times. The trader who waited for structure was never in the trade at all, because the structure never changed. Their divergence observations were still useful: they told them the trend was maturing, which is a reason to tighten management of existing longs. They just were not a reason to sell.

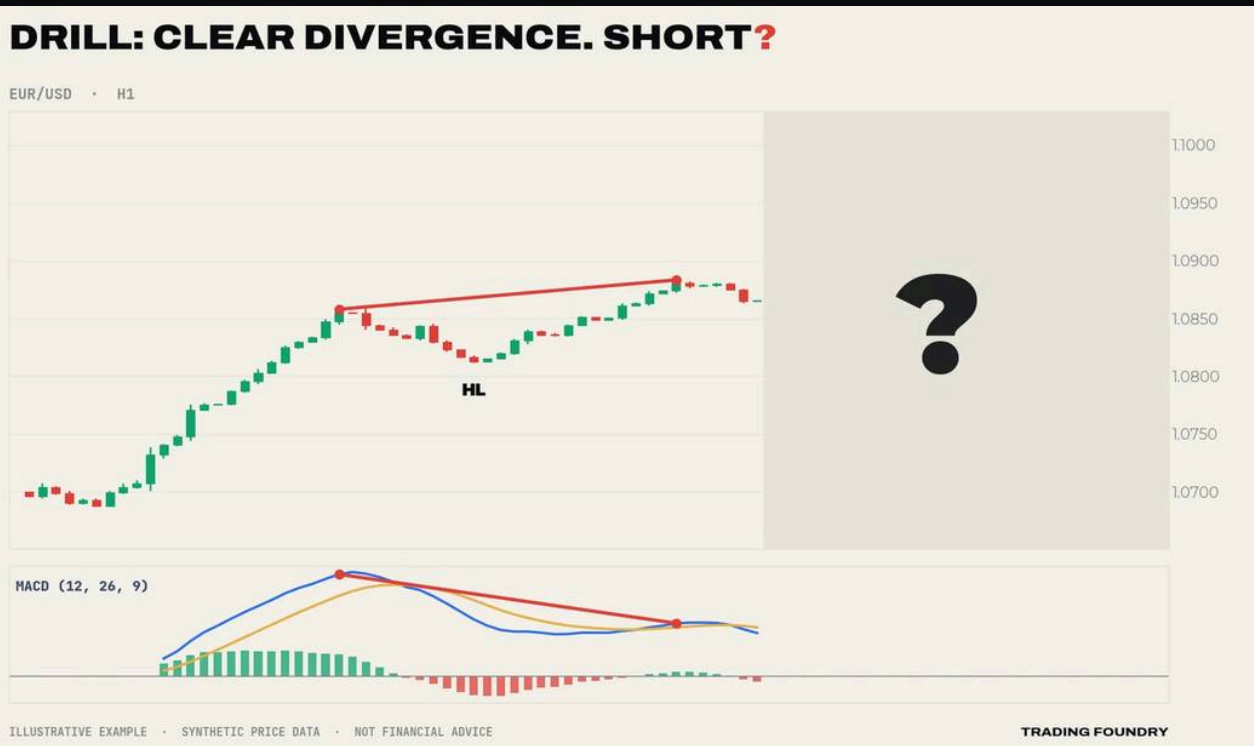
COMMON MISTAKE

Selling every bearish divergence in an uptrend. In strong trends divergence repeats, because speed naturally falls from its early peak. Without a lower high and a broken swing low, there is no reversal to trade.

Practice drill

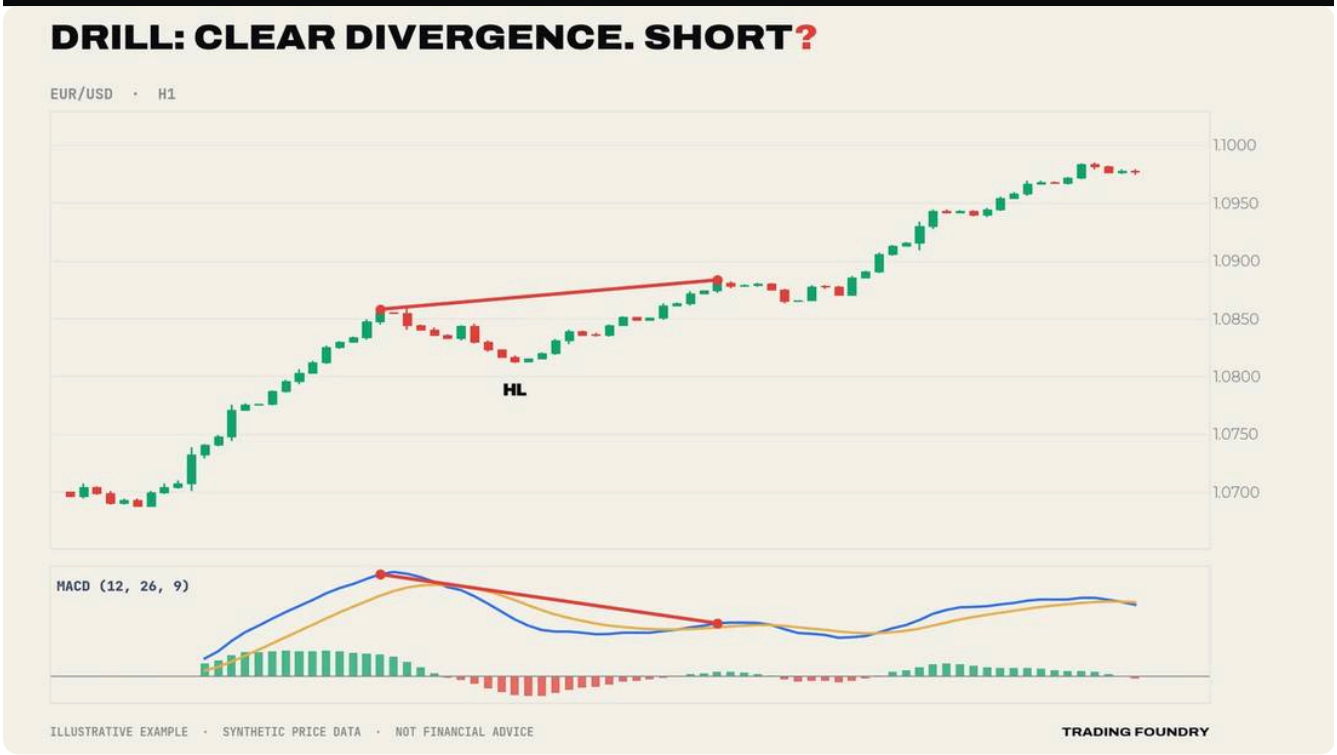
The chart below stops shortly after a clear bearish divergence. Price has started to drop. Decide whether there is a trade before you reveal.

PRACTICE DRILL · BUY, SELL OR WAIT? DECIDE, THEN TURN THE PAGE



A higher high on price, a much lower high on MACD, and price is dropping away from the peak. Short now, or not?

THE REVEAL



THE ANSWER

No trade. The divergence is genuine, but there is no lower high and no broken swing low — only a dip from the peak. The last higher low is still about 55 pips below current price. The confirmation rule says wait, and waiting was right: the dip held, price made a fresh high and the trend continued. The divergence marked a pause, not a top.

Line or histogram?

You will see divergence drawn on the histogram as often as on the MACD line, and the two can disagree. The histogram compares the MACD line with its own recent average, so it peaks earlier and more often; almost every second push in any trend produces a smaller histogram cluster, as Lesson 5 showed. That makes histogram divergence common and individually weak. The MACD line moves more slowly, and its peaks line up better with the swings you can see on price. For the confirmation rule in this lesson, use the line. If you also notice histogram divergence, treat it as background colour, not as a second, independent warning — both are built from the same averages, so two of them do not double the evidence.

Why divergence happens — honestly

You will read that divergence shows "smart money quietly selling into strength" or "institutions distributing". Perhaps sometimes. But the chart cannot show you anyone's intentions. What the chart shows is narrower and more reliable: the second push covered less distance per candle than the first. The MACD line, being a speed reading, fell.

There are dull and perfectly healthy reasons for that. Trends often start with a sharp repricing and settle into a steadier pace. A second push may happen in a quieter session. The first push may have been inflated by a news candle — the 'One candle, the biggest reading on the chart' example in Lesson 3 is a divergence waiting to happen. Price first. Indicator second. Stories, if at all, last.

Going deeper

Choosing the right peaks. Divergence must compare price swings with the MACD readings *at those swings*. Draw it between two clear swing highs on price, then check the MACD line at those same candles (or within a candle or two). Comparing a price high with an unrelated MACD peak several candles away is a common way to see divergences that are not there. The best divergences are between swings that are clearly separated — at least 10 to 15 candles apart — with the MACD line having pulled back meaningfully towards its signal line or zero between them.

Hidden divergence. The reverse pattern — price makes a **higher low** while the MACD line makes a **lower low** — is called hidden bullish divergence. It means a pullback pushed the speed reading lower than the previous pullback did, yet price held a higher low. Some traders treat this as a continuation signal in an uptrend. It is less reliable than the textbook claims suggest, and it overlaps heavily with the pullback methods in Lesson 7. Treat it as supporting evidence for a with-trend setup, not as a setup by itself.

Divergence and the zero line. Bearish divergences that form while the MACD line stays high above zero are weaker warnings than those that form after the line has already fallen back to zero between the peaks. In the second case, the market has already shown a full loss of trend speed once; the second peak is a recovery that failed to match the first.

Expectancy, illustratively. Suppose, hypothetically, that confirmed divergence trades reached 2R slightly less than half the time and stopped out at −1R the rest. The expectancy would be positive but modest, and the strategy would regularly produce runs of several losses. These numbers are invented to show the shape of the maths, not a claim about how any method performs. The real figures for your market can only come from your own testing.

Multiple timeframes. A divergence on a lower timeframe inside a strong higher-timeframe trend is usually just the end of a pullback or a push within that trend. Divergences that line up with a higher-timeframe level — a prior major high, a weekly range edge — carry more weight. Even then, the confirmation rule stays the same.

Summary

- Bearish divergence: higher high on price, lower high on the MACD line. Bullish: lower low on price, higher low on the MACD line.
- Divergence means the second push was slower. It is a warning of a maturing move, not a reversal signal.
- Trade it only after confirmation: a lower high (or higher low) and a close through the swing between.
- Strong trends produce repeated divergences. Without structural change, do not fade them.
- Charts show speed and structure, not intentions. Price first, indicator second.

CHECK YOURSELF · ANSWERS AT THE BACK

1. Which describes bearish MACD divergence?
   A. Lower high on price, higher high on MACD
   B. Higher high on price, lower high on the MACD line
   C. MACD crosses below zero
   D. Histogram turns red at a new high
2. In speed terms, what does bearish divergence tell you?
   A. Sellers are secretly in control
   B. The second push to a new high was slower than the first
   C. A reversal is guaranteed
   D. The trend has already ended
3. Under this lesson's confirmation rule, when is a bearish divergence short entered?
   A. At the second peak
   B. When the histogram turns red
   C. On a close below the swing low that formed after the second peak, once a lower high is in place
   D. When the MACD line crosses zero
4. In 'Two divergences, no reversal', why did the confirmation rule keep the trader out?
   A. The divergences were fake
   B. Price never made a lower high or broke a swing low
   C. The MACD line was below zero
   D. The timeframe was wrong
5. Where does the stop go on a confirmed bullish divergence long?
   A. Below the higher low that formed after the second price low
   B. At a fixed 10 pips
   C. Above the confirmation level
   D. There is no stop on divergence trades
6. A trader compares a price high with a MACD peak seven candles earlier and calls it divergence. What is wrong?
   A. Nothing
   B. Divergence must compare the MACD readings at the same swings as the price highs
   C. They should use RSI

D. Divergence only works on daily charts

The MACD Reset Pullback.

MODULE 3 — TRADING WITH MACD · LESSON 7

Trade with-trend pullbacks to the 20–50 EMA zone using the MACD reset towards zero, the histogram hook and a price trigger — with exact rules for entry, stop, target and when to skip.

Everything so far has been preparation. You know MACD measures speed (Lesson 1), that crossovers are late (Lesson 2), that the zero line tells you which side to trade (Lesson 4), and that the histogram hook marks the moment a counter-move stops gaining force (Lesson 5). This lesson puts those pieces together into one complete, rule-based method.

The idea is simple. Strong trends do not move in straight lines. They surge, pause, pull back, and surge again. The best place to join a trend is not during the surge — when MACD is high, the crossover is long gone and the stop would be miles away — but at the end of a pullback, when the trend has cooled off, price is back near its averages and the next surge has not yet begun. MACD is unusually good at showing that cooling-off. We will call it the **reset**.

The reset: what cooling off looks like

During an impulse, the MACD line climbs to a peak: the trend is fast. During the pullback that follows, the 12 EMA drifts back towards the 26 EMA and the MACD line falls back towards zero. When it gets close to zero without breaking through, the trend's short-term speed has been fully "spent" — yet the longer averages still slope upward and price is still above them. That combination is what a healthy pause looks like.



The impulse drives the MACD line to its peak. The pullback brings price back to the 20 EMA and the MACD line all the way back to just above zero, without crossing it. Price is still above a rising 50 EMA. The trend has cooled without breaking — the reset. The circled candle is where the pullback's selling ran out.

The reset solves two problems at once. It answers "has the pullback gone far enough?" — the speed built up in the impulse has been fully given back. And it answers "is the trend still alive?" — the MACD line did not cross zero, so the 12 EMA is still above the 26 EMA. A pullback that has not reset is probably not finished. A pullback that drives the MACD line deep below zero is probably not a pullback.

KEY POINT

The best pullback longs come after the MACD line has fallen back towards zero but not through it. Speed is spent; the trend is intact.

The complete rules

Here is the method for longs. Mirror everything for shorts.

1. Trend. Price is above the 50 EMA, the 20 EMA is above the 50 EMA, and both slope upward. The impulse before the pullback drove the MACD line to a clear positive peak.

2. Pullback. Price pulls back into the zone between the 20 and 50 EMAs. A wick below the 50 EMA is acceptable; a close below it cancels the setup.

3. Reset. The MACD line has fallen to less than half of its impulse peak and is still above zero.

4. Hook. The histogram prints its first clearly shorter red bar.

5. Trigger. Within three candles of the hook, a candle closes above the high of the previous candle. Enter at that close.

6. Stop. A few pips below the pullback's lowest low.

7. Target. 2R, or the prior swing high if it is closer and still at least 1.5R away. If the prior high gives less than 1.5R, skip the trade.

8. Risk. A fixed fraction of the account per trade — 0.5% to 1% is a common illustrative range. Position size is that risk divided by the stop distance.

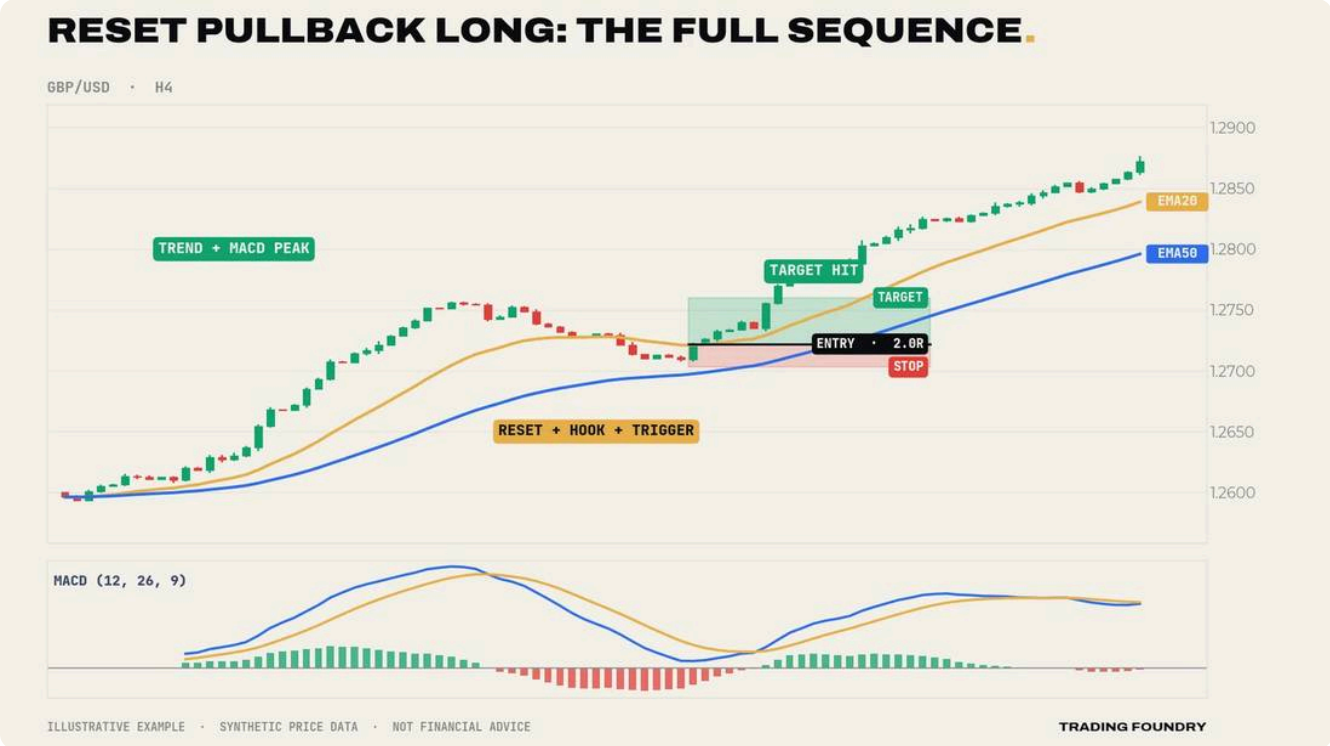
THE RULES

- Longs only when price is above a rising 50 EMA, with the 20 EMA above the 50 EMA (mirror for shorts)
- The pullback must reach the 20–50 EMA zone; a close beyond the 50 EMA cancels the setup
- The MACD line must have reset to below half its impulse peak while staying above zero
- Entry only on a price trigger (close beyond the previous candle's extreme) within three candles of a histogram hook
- Stop beyond the pullback extreme; target 2R or the prior swing if it offers at least 1.5R
- Risk a fixed fraction of the account; size = risk ÷ stop distance

Why the trigger must be a price candle

It is tempting to drop step five and enter on the hook itself, a few pips earlier. Resist it. The hook tells you the pullback has stopped accelerating; it does not tell you the pullback has stopped. Plenty of pullbacks slow, pause for a candle or two and then take another leg lower — the hook prints, the histogram even shrinks for a while, and price keeps falling. A close above the previous candle's high is different in kind. It is the first piece of evidence from price itself that buyers have taken control, however briefly. It also gives you a clean, objective entry that anyone could reproduce on the same chart. MACD tells you when to start watching. Price tells you when to act.

A winning trade

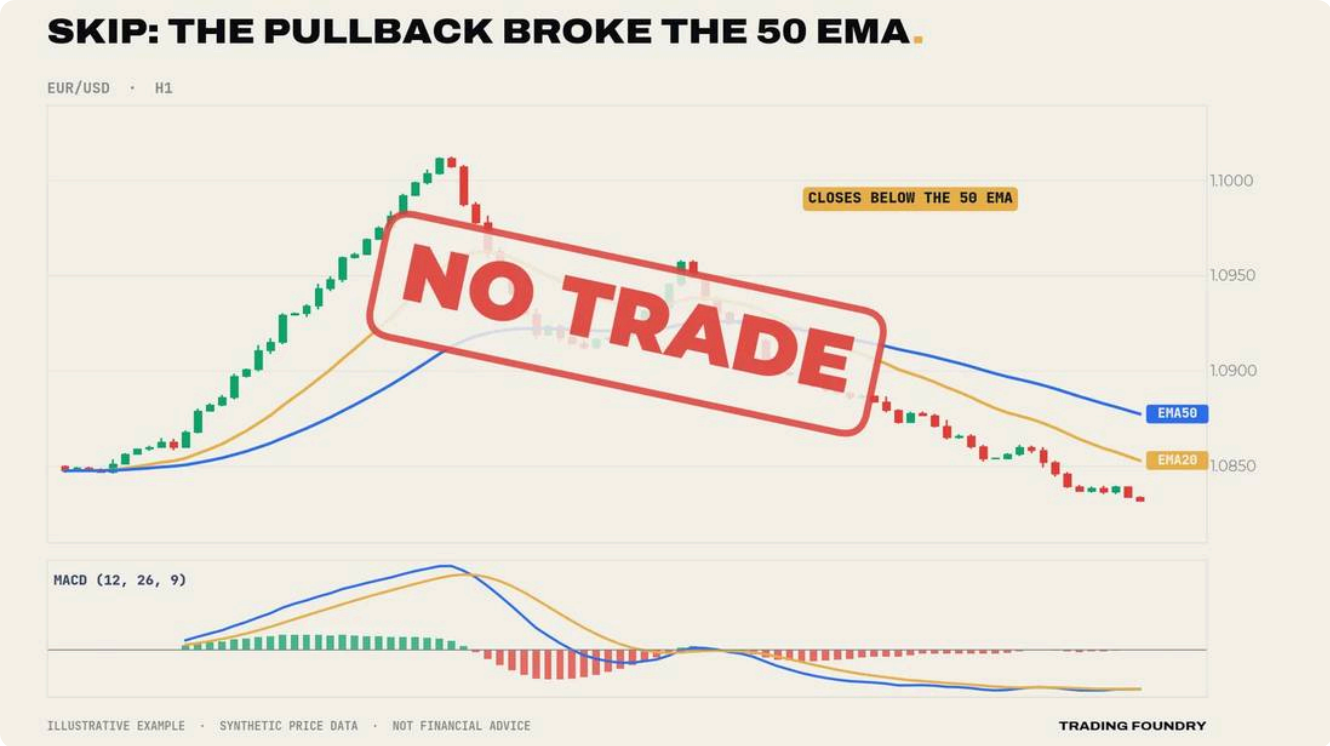


The uptrend pulls back into the 20–50 EMA zone without closing below the 50. The MACD line falls from its impulse peak to just above zero. The histogram hooks, and three candles later a candle closes above the previous high. Long at 122, stop under the pullback low at 103 (19 pips), target 2R at 160. Reached within ten candles.

Step through it with the rules in hand. Trend: price above both EMAs, stacked and rising. Pullback: into the zone, no close below the 50 EMA. Reset: the MACD line fell from about 34 at the peak to under 4 — well below half — and never crossed zero. Hook: the histogram's red bars peaked and began shrinking. Trigger: a strong candle closed above the previous candle's high. Every condition was met, so the trade was taken. With 19 pips of risk at an illustrative 1% per trade, the target paid 2%.

A trade that should have been skipped

The rules exist mostly for the trades they refuse. Here is a pullback that looks inviting — a big rally, a sharp dip, a strong bounce — but breaks two rules before any trigger appears.



A 100-pip drop in ten candles, faster than the rally it corrected, with several closes below the 50 EMA and the MACD line pushed below zero. The bounce that followed looked strong, and the histogram hooked beautifully, but the setup was already cancelled. Price failed below the high and the decline resumed.

Notice how convincing the bounce was. The histogram turned green, the MACD line curled up, candles were large and bullish. A trader without rules might have bought it. The rules had already answered the question: once price closed below the 50 EMA and the MACD line went through zero, this was no longer a pullback in an uptrend. It was something else, and the method does not trade something else.

COMMON MISTAKE

Re-qualifying a cancelled setup because the bounce looks strong. Once the 50 EMA close or the MACD zero-line rule is broken, the setup is gone. A new trend has to build its own structure first.

A valid trade that lost

Rules do not make losing trades disappear. They make them small and affordable. This one met every condition and still failed.



Trend, pullback, reset, hook and trigger all line up, so the long is taken at 119 with the stop at 104. Price rises only 11 pips before rolling over, slices back through the pullback low and keeps going. The stop turns an unknowable outcome into a known one: a loss of exactly 1R.

Could anything have warned you? Honestly, not much. The bounce stalled under the midpoint of the prior impulse and the MACD line barely turned up — but those observations are clearer in hindsight than they were at the time. This is the trade every method produces, and the reason risk per trade must be small and fixed. Over a series of trades, a method like this only needs its winners to be large enough and frequent enough to outweigh losses like this one. Whether they are, on your market and timeframe, is a question only your own testing can answer.

The short side

The method is symmetrical. In a downtrend, price below a falling 50 EMA, the MACD line has made a clear negative trough; the pullback rallies into the 20–50 EMA zone; the MACD line rises back towards zero without crossing it; the histogram hooks; a candle closes below the previous candle's low.



The downtrend's rally reaches the 20 EMA, short of the 50. The MACD line climbs from about −35 to just below zero. The histogram hooks and a candle closes below the previous low. Short at 135, stop above the rally high at 155, target 2R at 95. Price falls through the target.

Going deeper

Why the 20–50 EMA zone. The zone is a practical compromise. Pullbacks in strong trends often stop near the 20 EMA; pullbacks in steadier trends often reach the 50. Requiring the 20 alone would miss the deeper ones; requiring the 50 alone would miss the strongest trends entirely. The 50 EMA close rule is what separates "deep pullback" from "trend failure".

The half-peak reset threshold. The rule that the MACD line must fall below half its impulse peak is deliberately loose. Its purpose is to stop you buying a shallow dip while the trend is still very extended. If you test the method, try thresholds of a third and two-thirds as well. If results change dramatically between them, the method is fragile on that market. Robust methods tolerate sensible variation in their thresholds.

Higher-timeframe alignment. A refinement many traders use: only take trading-timeframe setups when the next timeframe up also has its MACD line above zero (for longs). In a variant of this method, when the higher-timeframe MACD is strongly positive, the trading-timeframe MACD line may be allowed to dip slightly below zero during the pullback. Choose one version, write it down and test it; mixing them trade by trade turns rules into opinions.

Trade management. The rules above use a fixed 2R target. Alternatives include taking half off at 1R and trailing the remainder below each new higher low, or exiting on a close below the 20 EMA. Each changes the shape of the results: fixed targets give more consistent outcomes; trailing exits occasionally catch large trends but give back more on average. There is no universally correct choice, only one that you test and apply consistently.

What breaks the method. Ranges (Lesson 8) and the late stage of trends, where pullbacks deepen and pushes shorten (Lesson 5). If three consecutive setups on a market fail, check whether the market is still trending at all before taking a fourth.

Summary

- The reset: during a pullback the MACD line falls back towards zero without crossing it — speed spent, trend intact.
- Rules: trend, pullback into the 20–50 EMA zone, reset below half the impulse peak, histogram hook, price trigger, stop beyond the pullback extreme, 2R target.
- A close beyond the 50 EMA or the MACD line through zero cancels the setup, however strong the bounce looks.
- Valid setups still lose. Fixed, small risk makes those losses affordable.
- The method is symmetrical for shorts.

PRE-TRADE CHECKLIST

☐ Price above a rising 50 EMA, 20 EMA above the 50 (mirror for shorts)

☐ The impulse drove the MACD line to a clear peak

☐ Pullback has reached the 20–50 EMA zone with no close beyond the 50 EMA

☐ MACD line has reset below half its impulse peak and is still on the trend side of zero

☐ Histogram has hooked

☐ Trigger candle has closed beyond the previous candle's extreme within three candles of the hook

☐ Stop placed beyond the pullback extreme; target at 2R or prior swing (minimum 1.5R)

☐ Position sized so the stop equals the planned fixed risk

CHECK YOURSELF · ANSWERS AT THE BACK

1. What is the MACD 'reset' in this method?

   A. The MACD line crossing zero

   B. The MACD line falling back towards zero during a pullback without crossing it

   C. Resetting the indicator settings to default

   D. The histogram turning green

2. During a long setup, price closes below the 50 EMA but then bounces strongly with a histogram hook. What do the rules say?

   A. Take the trade — the hook confirms it

   B. The setup was cancelled by the close below the 50 EMA; no trade

   C. Take it with half size

   D. Wait for the MACD zero cross and then buy

3. The impulse took the MACD line to +40. During the pullback it has fallen to +28. Has it reset?

   A. Yes

   B. No — it has not yet fallen below half its peak (+20)

   C. Yes, because it is above zero

   D. Only if the histogram is red

4. Where does the stop go on a reset pullback long?

   A. Below the 20 EMA

   B. A few pips below the pullback's lowest low

   C. At a fixed 50 pips

   D. Below the impulse's starting point

5. The prior swing high offers only 1.1R from your planned entry. What do the rules say?

   A. Take the trade with a 2R target anyway

   B. Skip — the target must offer at least 1.5R

   C. Move the stop closer to make it 2R

   D. Double the position size

6. In 'Valid setup, stopped out', what was the correct response to the loss?

   A. Change the rules immediately

   B. Accept it — a valid setup lost 1R, which is part of any method

   C. Re-enter immediately at a lower price

D. Remove stops from future trades

MACD in Ranges — Where It Whipsaws.

MODULE 3 — TRADING WITH MACD · LESSON 8

Recognise the sideways conditions that turn MACD into a whipsaw machine, switch your trend methods off in time, and use the histogram hook carefully at range edges instead.

Every trend tool has a natural enemy, and for MACD it is the sideways market. In a trend, MACD's lag is a manageable cost. In a range, it is fatal. The averages wind around each other, the MACD line oscillates around zero, and every crossover arrives just as price reaches the opposite side of the range. A crossover trader in a range is a person who buys every high and sells every low, perfectly, on schedule.

Markets spend a large share of their time going sideways — on many pairs and timeframes, more time than they spend trending. So this is not a footnote. Recognising a range early is worth more to a MACD trader than any refinement to their entry rules. This lesson shows you what a range does to MACD, the symptoms that tell you to switch your trend methods off, and a cautious way to use the histogram at range edges instead.

What a range does to MACD, mechanically

In a range, price returns to the same levels again and again. The 12 and 26 EMAs both settle near the middle of the range, so their difference — the MACD line — hovers around zero. Each swing from one side of the range to the other pushes the MACD line a little above zero, then a little below.

Now add the lag from Lesson 2. A swing from the bottom of the range to the top takes, say, eight candles. A signal-line crossover needs several candles to confirm. By the time a bullish crossover prints, the swing is half over. By the time the bearish crossover follows, price has already turned down from the top and is heading back to the middle. The result is a machine that buys in the middle on the way up and sells in the middle on the way down.



A 55-pip range and a stop-and-reverse crossover system. Every buy (B) printed halfway up a rally; every sell (S) printed halfway down a decline. Seven completed trades, seven losses, roughly 105 pips lost in total before spreads — on a market that went nowhere. The 50 EMA, flat as a table, was the warning.

Look at the timing. Each crossover is perfectly correct about the swing that has just happened — and each arrives too late to profit from it. There is nothing wrong with the indicator. It is measuring speed in a market where speed reverses every eight candles, which is roughly the delay the indicator needs to confirm anything.

KEY POINT

In a range, MACD's lag is about as long as the swings themselves, so every crossover arrives just in time to be wrong.

The symptoms: how to know you are in a range

You rarely know a range has started until it is a few swings old. The aim is to notice by the second or third swing, not the seventh. Four symptoms, in rough order of how early they appear:

1. **MACD peaks shrink towards zero.** As a trend fades into a range, each swing produces a smaller MACD peak, until the line is oscillating in a narrow band around zero.
2. **The MACD line crosses zero repeatedly.** Three or more zero-line crosses in the last 50 candles is a practical alarm.
3. **The 50 EMA flattens** and price starts crossing it in both directions.
4. **Price swings repeat between the same levels.** Highs fail at a similar price; lows hold at a similar price.



The rally ends at about 170 and the market starts swinging between two levels. Each swing produces a smaller MACD peak. By the right-hand side the MACD line has crossed zero five times in under thirty candles and the 50 EMA has flattened. Every symptom is present: switch the trend methods off.

When you see these symptoms, the reset pullback method from Lesson 7 must be switched off. The reset depends on an impulse that drove the MACD line to a clear peak; in a range there is no such impulse, so there is nothing to reset from. A dip towards the 20 EMA in a range is not a pullback. It is just the next swing.

COMMON MISTAKE

Reading a range as "a trend with deep pullbacks" and continuing to take trend setups. If the MACD line is crossing zero every few swings, the trend method has nothing to work with.

What to do while you wait

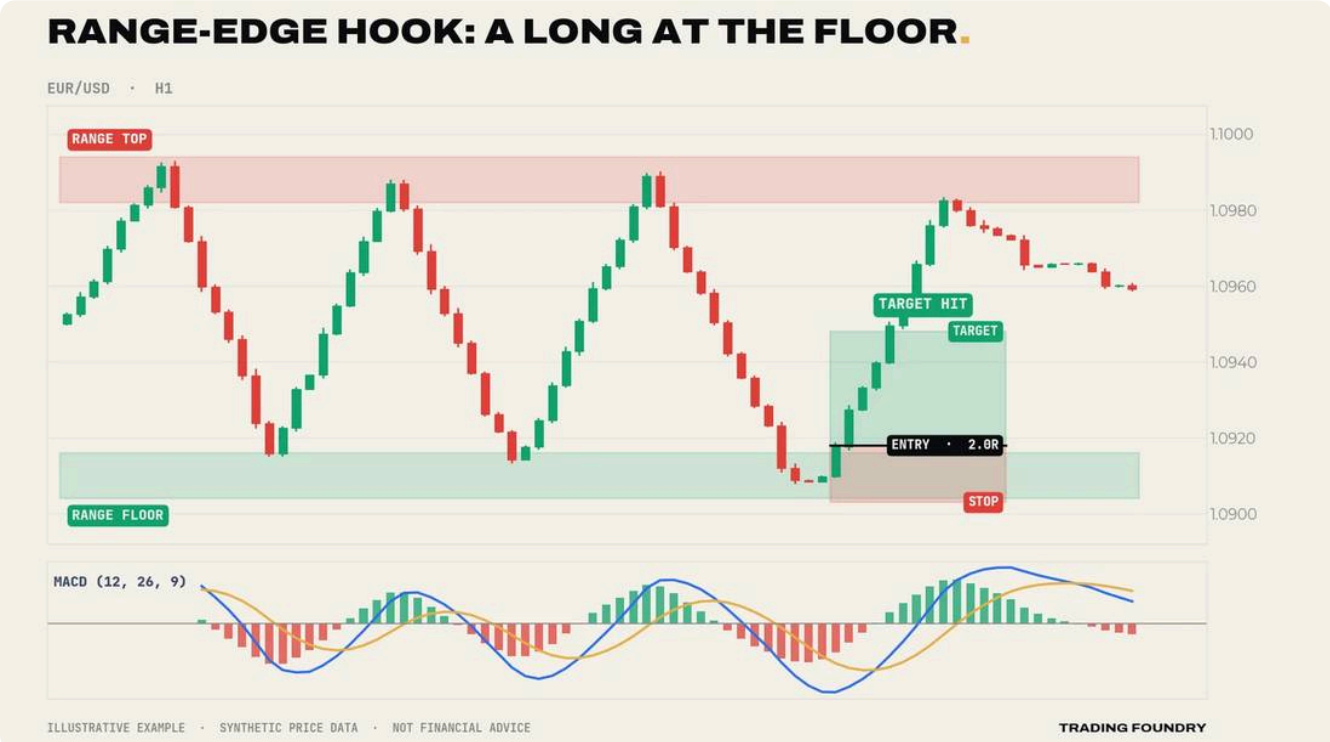
Switching the trend methods off does not mean switching the screen off. A range is the setup period for the next trend, and there are useful jobs to do. Mark the range edges clearly and note how many touches each has had. Watch the MACD band around zero: a band that is narrowing often precedes a break. Check the higher timeframe, because a range on H1 is frequently just a pause inside an H4 trend, and the H4 MACD will tell you which edge is more likely to give way. Above all, decide in advance what a break would look like — a close beyond the edge, the MACD line leaving its band — so that when it happens you recognise the new regime instead of arguing with it.

The range-edge hook

There is a way to use MACD in a well-defined range, but it is a different method with different rules, and it is more fragile than the trend method. The idea: instead of using the histogram hook to time the end of a pullback in a trend, use it to time the end of a swing into the edge of a range.

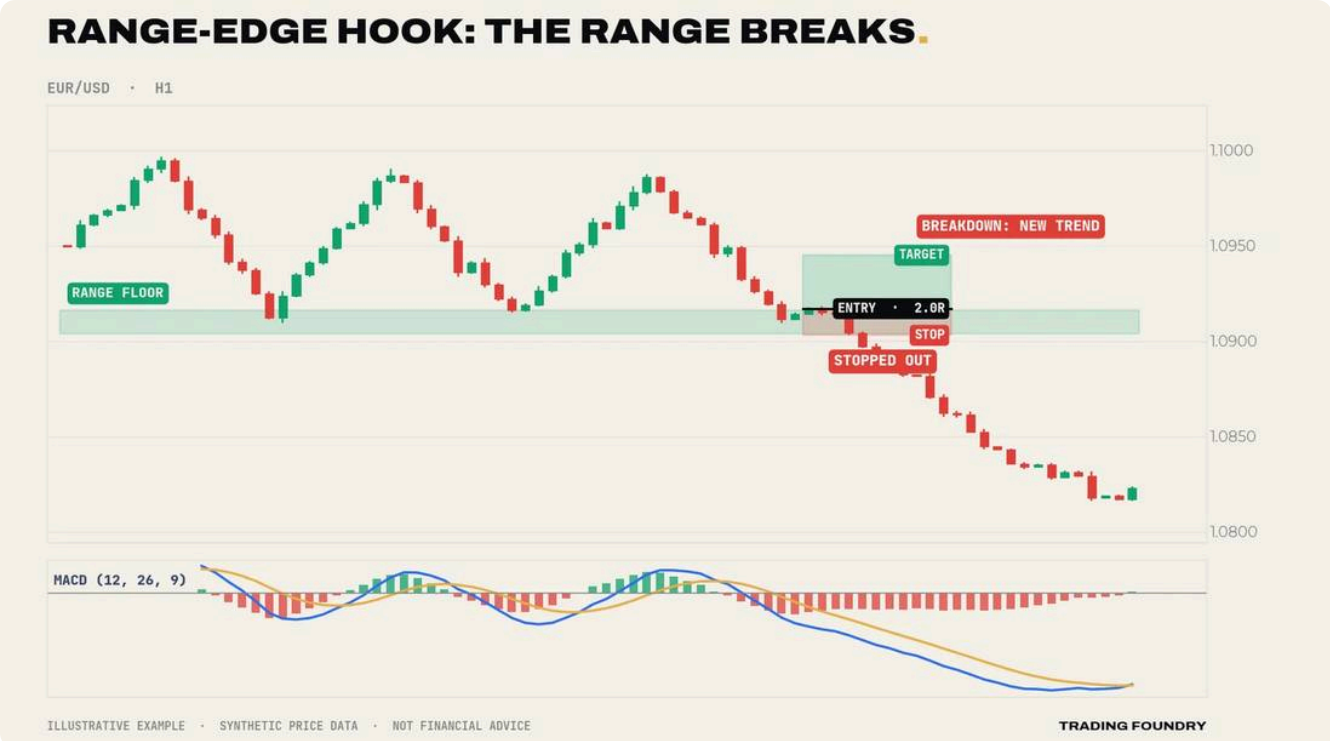
The rules for a long at the range floor (mirror for shorts at the top):

1. The range is established: at least two touches of both the floor and the top, and the range symptoms above are present.
2. Price swings down into the floor zone.
3. The histogram hooks (first clearly shorter red bar).
4. A candle closes above the previous candle's high, inside or just above the floor zone.
5. Stop a few pips below the floor zone.
6. Target 2R or the middle of the range, whichever is nearer — but skip the trade if the range midpoint offers less than 1.5R.



A range with three clean touches of each edge. Price swings down into the floor, the histogram hooks and a candle closes above the previous high. Long at 18, stop below the floor at 3, target 2R at 48 — just below the range midpoint. The swing back up reaches the target within a few candles.

Ranges end, and when they end they usually end through one of the edges you are trading from. That makes the range-edge hook a method with a built-in failure mode: the breakout. The stop below the floor is what keeps that failure affordable.



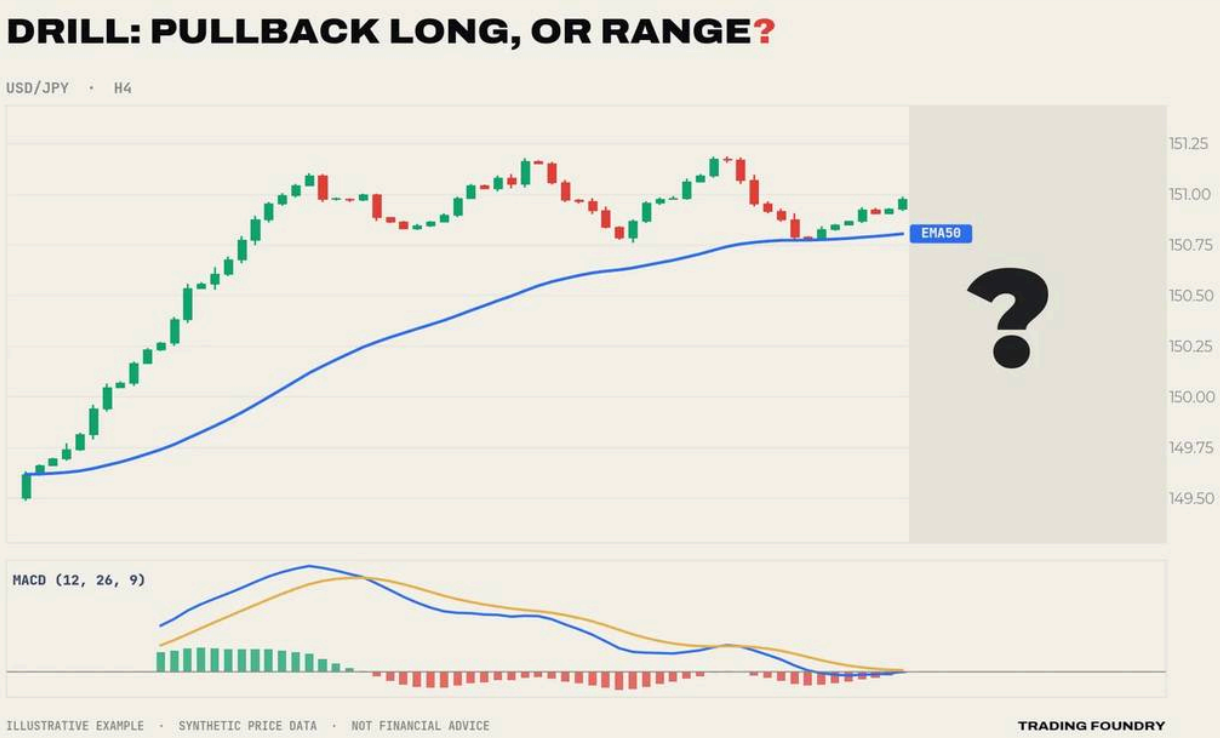
The same setup at the same floor — hook, trigger, stop below the zone. This time the bounce is feeble and price breaks through the floor a few candles later. The stop limits the loss to 1R. Notice the MACD line then drops well below zero and stays there: the range is over, and the trend methods are back in play — for shorts.

The losing trade carries useful information. When a range breaks, the MACD line typically leaves its narrow band around zero and moves decisively to one side. That is the first sign that a new trend is forming. The trend method does not fire immediately — it needs an impulse peak and then a pullback reset — but the range method should now be switched off.

Practice drill

This market was trending. Price has just bounced off a dip towards the 50 EMA and a candle has closed above the previous high, with the histogram contracting. Would you take a Lesson 7 reset long?

PRACTICE DRILL · BUY, SELL OR WAIT? DECIDE, THEN TURN THE PAGE



Price has bounced from about 128 and closed above the previous candle's high. The histogram is contracting. Is this a reset pullback long?

THE REVEAL



THE ANSWER

No trade. Three highs have failed near 160–165 and three lows have held near 128: a range. Each MACD peak is smaller than the last, the 50 EMA has flattened, and the MACD line has already crossed below zero — which cancels a reset setup on its own. Price rolled back below 128 and the stop would have been hit.

Going deeper

Why ranges are so common. Trends need a persistent imbalance — new information, flows that keep arriving in one direction. Most of the time, neither side has that edge, and price oscillates around a level that both sides find acceptable. On lower timeframes especially, ranges and choppy, overlapping price action can dominate. A trend method that sits out these periods is not missing opportunities; it is avoiding the conditions in which it has no edge.

A quantitative range filter. If you prefer a rule to judgement, combine two measures. First, count MACD zero-line crosses over the last 50 candles; three or more is a warning. Second, measure the 50 EMA's slope: the change in the 50 EMA over the last 20 candles, divided by the average candle range over the same period. When that ratio is small — for example below about 1 — the average is effectively flat. These thresholds are starting points for testing, not truths.

Range-edge versus trend method: different statistics. Range-edge trades tend to have smaller targets, because a range has a ceiling. They also fail by breakout, which tends to happen suddenly. Treat them as a separate method with its own journal, its own results and its own risk setting — many traders use half their normal risk on range trades. Do not let a run of range wins change how you trade trends, or vice versa.

Transitions are the hardest part. The most expensive moments are the two transitions: trend into range (when trend setups keep failing) and range into trend (when range-edge trades get run over). The symptom lists are your defence. A practical habit: after two consecutive losses with either method, stop and re-classify the market from scratch before the next trade.

Volatility compression. Ranges often narrow before they break. If the range's swings are getting smaller and the MACD band around zero is getting tighter, the breakout may be close. That is a reason to stop taking range-edge trades, not to guess the breakout direction.

Summary

- In a range the MACD line hovers around zero and crossovers arrive just as each swing ends — a whipsaw machine.
- Symptoms: shrinking MACD peaks, repeated zero-line crosses, a flat 50 EMA, repeated swing levels.
- When the symptoms appear, switch the trend methods off. There is no impulse to reset from.
- The range-edge hook can work in an established range, with a stop beyond the edge and a modest target.
- Ranges end through their edges. A MACD line that leaves its band around zero is the first sign of a new trend.

PRE-TRADE CHECKLIST

- ☐ At least two touches of both range edges
- ☐ Range symptoms present: MACD hugging zero, repeated zero crosses, flat 50 EMA
- ☐ Price in the edge zone, histogram hooked, trigger candle closed back inside
- ☐ Stop beyond the edge zone; target 2R or range midpoint (minimum 1.5R)
- ☐ Reduced risk compared with trend trades, if that is your plan

CHECK YOURSELF · ANSWERS AT THE BACK

1. Why do MACD crossovers perform so badly in ranges?
   A. Brokers widen spreads in ranges
   B. The indicator's confirmation delay is about as long as each swing, so signals arrive as swings end
   C. MACD is not calculated in ranges
   D. Ranges only occur at weekends
2. Which is the earliest typical symptom of a trend fading into a range?
   A. A news spike
   B. MACD peaks shrinking towards zero with each swing
   C. RSI above 70
   D. A 200 EMA crossover
3. The MACD line has crossed zero four times in the last 40 candles and the 50 EMA is flat. A dip reaches the 20 EMA with a histogram hook. Take a Lesson 7 reset long?
   A. Yes, every condition for a hook is met
   B. No — the trend method is switched off in a range; there is no impulse to reset from
   C. Yes, with double size
   D. Only if it is the London session
4. Where does the stop go on a range-edge long at the floor?
   A. At the range midpoint
   B. A few pips below the floor zone
   C. Above the range top
   D. No stop is needed in a range
5. After a range-edge long is stopped out, the MACD line moves well below zero and stays there. What does that suggest?
   A. The range is intact; buy again
   B. The range has likely broken into a new downtrend; switch off range trades
   C. MACD is malfunctioning
   D. Nothing
6. In 'Seven crossover trades, seven losses', what was visible on the price chart all along?
   A. A steep 50 EMA
   B. A flat 50 EMA and repeated swing levels

C. A head and shoulders

D. A news candle

Practical MACD Rule Sets.

MODULE 3 — TRADING WITH MACD · LESSON 9

Classify the market regime first, choose the matching MACD rule set, align it with the higher timeframe, and manage risk and open trades with written rules you can test.

You now have three MACD methods: the reset pullback for trends (Lesson 7), confirmed divergence for reversals (Lesson 6), and the range-edge hook for sideways markets (Lesson 8). Each one works in its own conditions and fails, sometimes badly, outside them. The single most common way traders destroy a sound method is not bad entries. It is using the right method in the wrong market.

This lesson turns the course into a working process. You will learn to classify the market before you look for any setup, choose the one rule set that fits, check it against the higher timeframe, and manage risk and open positions with rules rather than feelings. Nothing new is invented here. The value is in the order and the discipline.

Step one: classify the regime

Before you look at a single entry signal, answer one question: what kind of market is this? There are three answers, and each maps to one rule set.

- **Trending.** Price above a rising 50 EMA (or below a falling one), swing structure stepping in one direction, the MACD line making clear peaks on the trend side of zero and rarely crossing it. Rule set **A: reset pullback**.
- **Ranging.** Flat 50 EMA, repeated swing levels, the MACD line hugging zero and crossing it repeatedly. Rule set **C: range-edge hook** — or no trading at all, if you prefer.
- **Transitioning.** A mature trend showing divergence and fading pushes, or a range starting to break. Rule set **B: confirmed divergence**, but only once structure confirms; until then, stand aside.



The same market in three regimes. An uptrend with clear MACD peaks above zero (rule set A). A sideways phase with shrinking MACD swings and a flattening 50 EMA (rule set C, or stand aside). Then a breakdown, the MACD line crossing zero and staying below it — a new downtrend, and rule set A again, for shorts. The gaps between the shaded areas are transitions: wait.

The unshaded gaps in that chart matter as much as the shaded areas. Between regimes there is a period when the old method has stopped working and the new one has not started. The correct action in that gap is usually nothing. A trader who is comfortable doing nothing for a while has a large advantage over one who needs a trade every session.

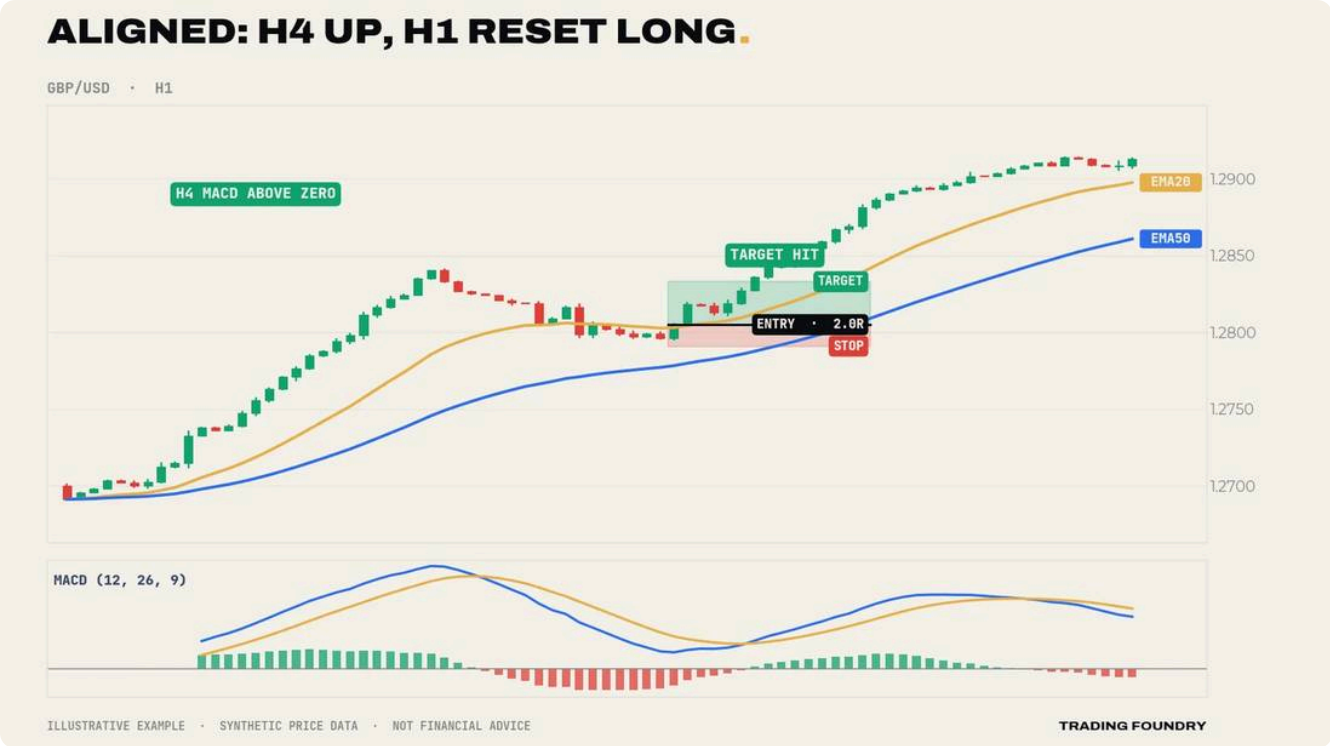
KEY POINT

Classify first, then choose the rule set. Never look for an entry before you have decided what kind of market you are in.

Step two: align with the higher timeframe

Lesson 3 introduced the principle: direction from the higher timeframe, timing from the lower one. In rule form, for rule set A:

- **Longs** on the trading timeframe only when the next timeframe up (roughly four to six times longer) has its MACD line above zero.
- **Shorts** only when it is below zero.
- If the higher timeframe is in a range (MACD hugging zero), trade the trading timeframe as a range or stand aside.



The four-hour MACD line is above zero, so H1 longs are permitted. The H1 pullback reaches the 20–50 EMA zone, the MACD line resets from 32 to about 5, the histogram hooks and a candle closes above the previous high. Long at 105, stop 91, target 2R at 133, reached six candles later.

Now the same H1 picture, but with the higher timeframe pointing the other way.



On its own, this H1 chart offers a textbook reset long: trend, pullback, reset, hook, trigger. But the four-hour MACD line is below zero — this H1 'uptrend' is a rally inside a higher-timeframe downtrend. The alignment rule says skip. A trader who took it anyway was stopped out as the larger trend reasserted itself.

The second chart is the best argument for the alignment rule. Nothing on the H1 chart was wrong. Every condition of rule set A was met. The information that should have stopped the trade was simply not on that chart. That is why classification and alignment come before setups: they put the setup in context.

Step three: the rule sets, written down

Here are the three methods in compressed form. They are the rules from Lessons 6, 7 and 8, written as you might keep them on a card next to your screen.

THE RULES

- A (Reset pullback, trending regime): higher-timeframe MACD on your side of zero; price beyond a sloping 50 EMA; pullback into the 20–50 EMA zone; MACD line reset below half its impulse peak without crossing zero; histogram hook; trigger close beyond previous candle; stop beyond pullback extreme; 2R target
- B (Confirmed divergence, transition): divergence between two clear swings on the MACD line; a lower high (or higher low) forms; trigger on a close through the swing between; stop beyond that lower high (or higher low); 2R target or next level (minimum 1.5R)
- C (Range-edge hook, ranging regime): two or more touches of each edge; range symptoms present; hook and trigger at the edge; stop beyond the edge zone; target 2R or range midpoint (minimum 1.5R); reduced risk
- In every rule set: no trade without a stop placed at entry, and no stop moved further away once placed

Notice what is *not* on the card. There is no plain crossover method, because Lessons 2 and 8 showed it has no structural stop and no defence against ranges. There is no zero-line-cross entry, because it is too late. There is no "divergence alone" entry, because Lesson 6 showed how it fails in strong trends. Each of those ideas survives only as part of something else: the zero line as a filter, the crossover's cousin the hook as a timing cue, divergence as a warning.

Step four: risk rules

Entries decide whether a trade is good. Risk rules decide whether a bad run is survivable. The numbers below are illustrative starting points; choose your own and keep them fixed.

- **Risk per trade:** a fixed fraction of the account, for example 0.5% to 1%. Position size = risk ÷ stop distance.
- **Correlated exposure:** EUR/USD and GBP/USD longs are largely the same bet on a weaker US dollar. Treat correlated positions as one risk unit, or cap combined risk.
- **Daily and weekly limits:** for example, stop for the day after losses totalling 2R, and for the week after 5R. This protects you from the most dangerous trader in the room — yourself after three losses.
- **No averaging down.** Adding to a losing position turns a 1R loss into a much larger one.

Run the numbers on a losing streak, which every method has. With 1% risk per trade, eight consecutive losses take the account down by roughly 7.7%. With 3% risk per trade, the same streak costs about 22%. The method has not changed; only the sizing has. Runs of that length are not rare in any method with a modest strike rate, so size for them.

Step five: managing open trades

MACD has one more job once you are in a trend trade: telling you when to become defensive.



A long from earlier in the trend is still open with part of the position. When the new high arrives with a lower MACD peak, the trader stops adding and moves the stop for the remainder to just below the last higher low. The exit comes when price closes below it — not at the top, but well before most of the decline.

That is the correct use of divergence on an open position: not a reason to reverse, but a reason to protect. Two practical management rules:

1. **When a bearish divergence forms against an open long, take no new longs and trail the stop for any remaining position below the most recent higher low.** Mirror for shorts.
2. **When the MACD line crosses zero against an open position that has not yet hit its target, exit.** The trend filter has closed; the reason for the trade is gone.

COMMON MISTAKE

Changing the rules mid-trade. Moving a target further away because "momentum is strong", or a stop further away because "it's just a pullback", replaces a tested process with a feeling. Write the management rules before the trade and follow them.

Step six: test before you trust

Every rule in this course is a hypothesis about how a market behaves, not a fact. Before trading any rule set with real money, test it:

1. Pick one market and one timeframe.
2. Scroll back at least a few hundred candles. Hide the future (most platforms have a replay mode).
3. Step forward a candle at a time. Classify the regime. When a setup appears, record entry, stop and target before you look further.
4. Record every trade — winners, losers and the ones you skipped and why.
5. After at least 50–100 trades, look at the results: average R per trade, longest losing streak, worst drawdown. Then decide whether the method deserves a small, live, demo-size trial.

Results from a careful manual test are still optimistic — you know the course's rules were written with hindsight, and live trading adds spreads, slippage and emotion. Treat any result as an upper bound, not a forecast.

Going deeper

Rules that a stranger could follow. A good test of your written rules is whether someone else could apply them to the same chart and reach the same decision. "Strong trend" fails that test; "price above a 50 EMA whose value is higher than 20 candles ago" passes it. Every place where your rules say "clear", "strong" or "good", either define it numerically or accept that you are using judgement and record it as such.

One method per regime, one change at a time. When results disappoint, change one rule, re-test, and compare. Changing three things at once makes it impossible to know which change helped. Keep a version history of your rules, with dates.

Expectancy, illustratively. A method's average result per trade is (win rate × average win) − (loss rate × average loss). With 2R targets and 1R stops, a hypothetical method that wins 40% of the time averages 0.4 × 2 − 0.6 × 1 = +0.2R per trade before costs; at 30% it averages −0.1R. These figures are invented to show the arithmetic, not to describe any real method. The point is how sensitive the outcome is to small changes in win rate, costs and average win — which is why testing matters.

When nothing fits. Some days no regime is clear. The market is neither trending cleanly nor ranging cleanly. The right rule set for those days is the empty one. Write that down too: "If I cannot classify the regime within two minutes, I do not trade it."

Summary

- Classify the regime first: trending (set A), ranging (set C), transitioning (set B after confirmation, otherwise wait).
- Align with the higher timeframe: its MACD zero line sets the permitted direction.

- Keep the three rule sets written down; plain crossovers, zero-cross entries and unconfirmed divergence are not on the card.
- Fix risk per trade, cap correlated exposure and set daily and weekly loss limits.
- Use divergence to protect open trades, and exit when the MACD line crosses zero against you. Test everything before trusting it.

PRE-TRADE CHECKLIST

☐ Regime classified: trending, ranging or transitioning (if unclear, no trade)

☐ Rule set chosen to match the regime

☐ Higher-timeframe MACD on the permitted side of zero (rule set A)

☐ Every condition of the chosen rule set met, with no exceptions

☐ Stop placed at entry; target at least 1.5R; position sized to fixed risk

☐ Correlated exposure and daily loss limit checked

☐ Management rules for divergence and zero-line exit written down before entry

CHECK YOURSELF · ANSWERS AT THE BACK

1. What is the first step before looking for any MACD entry?
   A. Check the histogram colour
   B. Classify the market regime
   C. Change the settings
   D. Look for divergence
2. The H1 chart shows a perfect reset long, but the H4 MACD line is below zero. What does the alignment rule say?
   A. Take it — the H1 setup is perfect
   B. Skip it — the higher timeframe does not permit longs
   C. Take it with a wider stop
   D. Switch to rule set C
3. Which of these is NOT one of the course's rule sets?
   A. Reset pullback
   B. Confirmed divergence
   C. Range-edge hook
   D. Plain signal-line crossover
4. With 1% risk per trade, roughly how much does a run of eight consecutive losses cost?
   A. About 2%
   B. About 7.7%
   C. About 22%
   D. About 50%
5. A bearish divergence forms while you hold part of a long position. What does the management rule say?
   A. Reverse to short immediately
   B. Take no new longs and trail the stop below the most recent higher low
   C. Add to the position
   D. Ignore it
6. Using the illustrative formula, what is the expectancy of a hypothetical method with a 40% win rate, 2R wins and 1R losses, before costs?
   A. −0.2R
   B. +0.2R

C. +0.8R

D. +1.0R

Capstone: The Complete MACD Playbook.

MODULE 4 — CAPSTONE · LESSON 10

Run the whole MACD process from regime to exit on three full worked trades, then use the strategy passport and pre-trade checklist to apply it on your own charts.

This is the lesson to keep open next to your charts. It gathers everything from the course into one process — the playbook — and then walks through three complete trades from first glance to final exit: two that worked and one that did not. It finishes with the strategy passport, the full pre-trade checklist and a final drill.

Before you start, one reminder about what this playbook is. It is a structured way of using MACD that respects what the indicator actually measures: speed and acceleration, built from averages, always late. It is not a prediction machine and it does not remove losing trades. Its job is to make sure that every trade you take has a reason, a stop and a size you decided in advance, and that you stay out of the conditions where MACD has nothing useful to say.

The playbook, step by step

Step 1 — Read the higher timeframe. Open the timeframe four to six times above your trading chart. Is its MACD line above zero, below zero, or hugging it? This sets your permitted direction. Hugging zero means no trend trades.

Step 2 — Classify the regime on your trading timeframe. Trending: price beyond a sloping 50 EMA, clear MACD peaks on one side of zero. Ranging: flat 50 EMA, repeated swing levels, MACD crossing zero repeatedly. Transitioning: divergence and fading pushes at the end of a trend, or a range edge giving way. If you cannot classify it in two minutes, stop here.

Step 3 — Choose the rule set. Trending and aligned with the higher timeframe: rule set A, the reset pullback. Transitioning with a clear divergence: rule set B, but only after structure confirms. Ranging: rule set C, the range-edge hook, or no trade.

Step 4 — Wait for every condition. For rule set A: pullback into the 20–50 EMA zone without a close beyond the 50; MACD line reset below half its impulse peak without crossing zero; histogram hook; trigger candle within three candles of the hook. Missing one condition means no trade.

Step 5 — Place the stop, then size the position. Stop beyond the pullback extreme (A), the lower high or higher low (B), or the range edge (C). Position size = fixed risk ÷ stop distance.

Step 6 — Check the target. 2R, or the nearest obstacle if it offers at least 1.5R. Less than 1.5R: skip.

Step 7 — Manage by rule. Exit at the target or stop. If a divergence forms against an open position, take no new trades in that direction and trail any remainder beyond the latest swing. If the MACD line crosses zero against an unfinished trade, exit.

Step 8 — Record it. Regime, rule set, entry, stop, target, outcome in R, and one line about anything you would do differently.

Worked trade 1: a reset pullback long



Higher timeframe up, H4 trending. The pullback reaches the 20–50 EMA zone; a wick tests the 50 EMA but the close holds above it; the MACD line resets from about 37 to 2 without crossing zero; the histogram hooks as a strong candle closes above the previous high. Long at 128, stop 102 under the pullback low, target 2R at 180. Hit nine candles later.

Walk through the playbook.

Higher timeframe: the daily MACD line is above zero, so longs are permitted. **Regime:** on the H4 chart price is above a rising 50 EMA, the 20 EMA is above the 50, and the impulse drove the MACD line to a clear peak of about 37 — trending, so rule set A. **Conditions:** the pullback dropped through the zone between the 20 and 50 EMAs and one candle's wick pierced the 50 — but that candle closed just above it, so the setup survived. The MACD line fell to about 2: well under half its peak, still above zero. On the very next candle the histogram's red bar shrank — the hook — and the same candle closed far above the previous high: the trigger. **Stop and size:** the pullback's low sat at 106, so the stop went at 102, 26 pips away. At an illustrative 1% risk, that sets the size. **Target:** 2R at 180. The prior high at 188 was further away, so it did not limit the target. **Outcome:** the target was reached nine candles later: +2R.

Notice what the trader did not do. They did not buy during the impulse because MACD was "strong". They did not buy the first red histogram bar. They did not wait for the signal-line cross, which came six candles later and over 30 pips higher. And they did not panic when a wick poked through the 50 EMA: the rule is about closes, and the close held. Each of those choices is a lesson from earlier in the course.

Worked trade 2: a confirmed divergence short

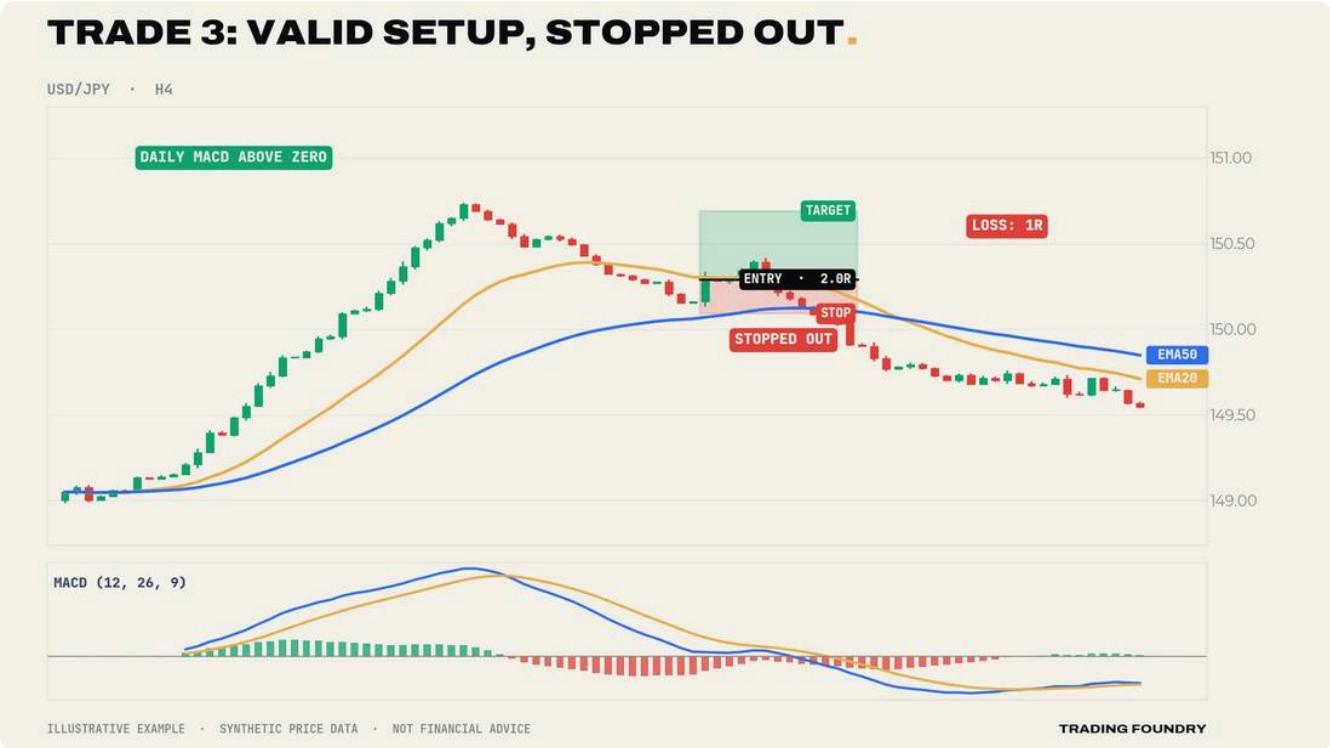


A new high with a much lower MACD peak: bearish divergence. The trader waits. Price drops to a swing low at 132, bounces to a lower high near 160, then closes below the swing low. Short at 130, stop above the lower high at 163, target 2R at 64. The decline reaches it eleven candles later.

Higher timeframe: the daily MACD line had been falling towards zero — the larger trend was losing speed, which is the context in which rule set B makes sense. **Regime:** a mature uptrend showing divergence: transitioning. **Conditions:** the divergence was clear — price's second peak was about 30 pips higher, the MACD line's second peak about half the height of the first. But the playbook says divergence is a warning, so the trader did nothing at the peak. Price fell to a swing low at 132, bounced to a lower high near 160 and then closed below 132. That close was the trigger. **Stop and size:** above the lower high, at 163: 33 pips. **Target:** 2R at 64. **Outcome:** reached: +2R.

The trade was taken roughly twenty candles after the divergence first appeared, and about 55 pips below the high. Someone who sold the high on divergence alone would have done better on this chart — and would have been run over on the 'Two divergences, no reversal' chart in Lesson 6. The confirmation rule gives up some trades' best prices in exchange for avoiding the worst ones.

Worked trade 3: a valid long that lost



Every condition is met: higher timeframe up, trend on H4, pullback into the EMA zone, MACD reset from about 35 to under 2, hook and trigger. Long at 129, stop 109, target 2R at 169. Price rises 12 pips, stalls and breaks the pullback low. The stop is hit for a loss of exactly 1R.

Higher timeframe: up. **Regime:** trending. **Conditions:** all met — the MACD line reset from about 35 to under 2, never crossed zero, the histogram hooked and a strong candle closed above the previous high. **Stop and size:** below the pullback low, 20 pips. **Outcome:** price rallied 12 pips, stalled and broke down. Stopped out: −1R.

What should the trader do with this? Record it, and nothing else. They should not change the rules because of one loss, should not add a filter that happens to exclude this exact chart, and should not double the size on the next trade to "win it back". Over a long series of trades, the method is judged by its average outcome in R, which includes losses like this. Across these three trades the net result was +3R — but three trades prove nothing in either direction, and the next three could easily be three losses.

KEY POINT

A valid trade that loses is not a mistake. A mistake is a trade taken without every condition, a stop moved, or a position sized by feel.

The strategy passport

The whole course on one card.

- **Market:** liquid major currency pairs and gold on H1 to D1, in a trending regime confirmed by the higher timeframe's MACD line.
- **Wait for:** a pullback into the 20–50 EMA zone that resets the MACD line to below half its impulse peak without crossing zero.
- **Trigger:** the first histogram hook, followed within three candles by a close beyond the previous candle's extreme.
- **Stop:** a few pips beyond the pullback's extreme. Size the position so the stop equals your fixed risk.
- **Target:** 2R, or the prior swing if closer and still at least 1.5R. Exit early if the MACD line crosses zero against you.
- **Avoid:** ranges (MACD hugging zero, flat 50 EMA), counter-trend crossovers, unconfirmed divergence, single-candle news spikes and any setup that conflicts with the higher timeframe.

The full pre-trade checklist

PRE-TRADE CHECKLIST

☐ Higher-timeframe MACD line on the permitted side of zero (not hugging it)

☐ Regime classified on the trading timeframe; rule set chosen to match

☐ Price beyond a sloping 50 EMA, with the 20 EMA on the same side (rule set A)

☐ Impulse drove the MACD line to a clear peak; pullback has reset it below half that peak without crossing zero

☐ Pullback reached the 20–50 EMA zone with no close beyond the 50 EMA

☐ Histogram has hooked; trigger candle closed beyond the previous candle's extreme within three candles

☐ Stop placed beyond the pullback extreme (or lower high / higher low for rule set B, range edge for C)

☐ Target at 2R, or nearest obstacle if at least 1.5R; otherwise skip

☐ Position size = fixed risk ÷ stop distance; correlated exposure and daily loss limit checked

☐ No major scheduled news due before the trigger candle closes

☐ Management plan written: divergence → protect; MACD zero-line cross against the trade → exit

THE RULES

- No trade without a classified regime and a matching rule set
- No trade against the higher-timeframe MACD zero line
- No entry without a price trigger; MACD alone never triggers a trade
- No trade without a stop placed at entry; never move a stop further away
- Fixed risk per trade; no averaging down; stop trading for the day at the daily loss limit

Final drill

The chart stops on the close of a candle in a downtrend. Work through the playbook — every step — before you press reveal. Assume the higher-timeframe MACD line is below zero.



Downtrend on H1, higher timeframe aligned. Price has rallied into the EMAs and the last candle has just closed. Trade or no trade?

THE REVEAL



THE ANSWER

Trade — a valid reset pullback short. Price is below a falling 50 EMA; the rally reached above the 20 EMA but stayed below the 50; the MACD line rose from about −35 to about −4, past half its trough without crossing zero; the histogram hooked two candles ago and this candle closed below the previous low. Short at 120, stop above the rally high at 137, target 2R at 86. The target was reached five candles later.

Going deeper

Building your own evidence. The worked trades in this course are training charts: illustrative, built to show the process clearly, not records of real market history. The only evidence that matters for your trading is the evidence you gather yourself. Run the step-by-step test from Lesson 9 on at least one market and timeframe, record at least 50 to 100 setups, and look honestly at the average R, the longest losing run and the deepest drawdown before risking money.

Adapting without breaking. You will want to adjust things: different targets, a trailing exit, a different EMA zone. That is fine, provided you change one thing at a time, test it, and keep the parts of the method that carry its logic — the regime classification, the higher-timeframe alignment, the price trigger and the structural stop. Those four are what turn MACD from a lagging line into a usable process.

What MACD will never do. It will never see support and resistance, never know about tomorrow's interest-rate decision and never tell you a move is about to start. Everything in this playbook is built around that honesty: price structure decides, MACD describes speed, and risk rules keep you in the game long enough for the process to matter.

Where to go next. The same speed-and-acceleration thinking applies to other momentum tools. RSI and the stochastic measure similar things on bounded scales; rate-of-change measures speed directly. If you understand why MACD behaves the way it does, you will understand them quickly — and you will be just as sceptical of their "signals".

Summary

- The playbook: higher timeframe, regime, rule set, every condition, stop then size, target check, rule-based management, record.
- Trade 1 (reset long) and trade 2 (confirmed divergence short) each returned +2R; trade 3, equally valid, lost 1R.
- A valid loss is not a mistake; broken rules are.
- The passport and checklist summarise the whole course for daily use.
- Evidence comes from your own testing. Training charts teach the process; they do not prove a result.

CHECK YOURSELF · ANSWERS AT THE BACK

1. What is step 1 of the playbook?

   A. Look for a histogram hook

   B. Read the higher timeframe's MACD line to set the permitted direction

   C. Place the stop

   D. Check for divergence

2. In Trade 1, why did the trader not wait for the signal-line crossover?

   A. Crossovers are never valid

   B. The hook plus price trigger came earlier, giving a cheaper entry and tighter stop; the crossover came over 30 pips higher

   C. The crossover was bearish

   D. The platform did not show it

3. In Trade 2, when was the short entered?

   A. At the second peak, on divergence

   B. On the first red histogram bar

   C. On the close below the swing low after a lower high had formed

   D. When the MACD line crossed zero

4. After Trade 3 lost 1R with every condition met, what is the correct response?

   A. Add a filter that would have excluded this chart

   B. Double the next position

   C. Record it and keep following the rules

   D. Stop using stops

5. According to the passport, which is on the 'avoid' list?

   A. Pullbacks into the 20–50 EMA zone

   B. Ranges where MACD hugs zero

   C. Histogram hooks

   D. 2R targets

6. Which of these is a mistake rather than an acceptable loss?

   A. A valid setup that hits its stop

   B. A trade taken with the higher-timeframe MACD line on the wrong side of zero

   C. A 2R target that is reached

D. Skipping a trade with only 1.2R to the prior high

Strategy passport.

The whole setup on one page. Print it and keep it by your screen.

MARKET

Liquid major FX pairs and gold, H1 to D1, in a trending regime confirmed by the higher timeframe's MACD line

WAIT FOR

A pullback into the 20–50 EMA zone that resets the MACD line below half its impulse peak without crossing zero

TRIGGER

The first histogram hook, then a close beyond the previous candle's extreme within three candles

STOP

A few pips beyond the pullback extreme; size the position so the stop equals a fixed 0.5–1% risk (illustrative)

TARGET

2R, or the prior swing if closer and at least 1.5R; exit early if the MACD line crosses zero against you

AVOID

Ranges where MACD hugs zero, counter-trend crossovers, unconfirmed divergence, news-spike readings and higher-timeframe conflict

Quiz answers.

Lesson 1, Q1: A. MACD line = 12 EMA minus 26 EMA. It contains no volume or candle-count information.

Lesson 1, Q2: B. Above zero means the fast EMA is still above the slow EMA; falling means the gap is narrowing — the uptrend is slowing. Price can still be rising.

Lesson 1, Q3: A. The histogram is MACD minus signal, so a bullish signal-line cross is the histogram moving from negative to positive.

Lesson 1, Q4: B. The 26 EMA lags about 7 candles more than the 12 EMA, so the gap settles near 7 × 4 = 28 pips. This shows the line is a speed reading.

Lesson 1, Q5: C. MACD is built from averages of closes and has no knowledge of levels. Levels come from the price chart.

Lesson 2, Q1: C. The histogram measures acceleration, so it turns as soon as the decline stops speeding up — before the signal cross and well before the zero cross.

Lesson 2, Q2: B. Position size is risk divided by stop distance. A later entry widens the stop distance, so the same risk buys fewer units.

Lesson 2, Q3: B. In a V reversal, the MACD's speed reading changes abruptly, so the cross arrives within a few candles. Slow, rounded turns delay it.

Lesson 2, Q4: B. A healthy pullback slows the trend enough to trigger a bearish cross. The system sold and then bought back higher when the trend resumed.

Lesson 2, Q5: B. When lag eats a large fraction of the typical move, the remaining reward is small relative to the stop distance.

Lesson 2, Q6: A. Both a pullback and a reversal reduce trend speed, so both trigger the opposite crossover. Only price structure separates them.

Lesson 3, Q1: A. Shorter averages respond sooner but also respond to small wiggles. Earlier always comes with noisier.

Lesson 3, Q2: C. Each H4 candle contains four H1 candles, so multiply each period by four.

Lesson 3, Q3: C. The MACD line stayed above zero throughout, while the signal line produced eight crossovers.

Lesson 3, Q4: B. A higher-timeframe pullback naturally appears as a small downtrend on the lower timeframe. The disagreement is information.

Lesson 3, Q5: B. MACD is not bounded like RSI. A reading is only high or low relative to the same market's recent history.

Lesson 3, Q6: B. MACD reacts to candle size, not significance. Let price confirm whether the move holds.

Lesson 4, Q1: B. Below zero the filter permits shorts only. A bullish cross there is a counter-trend signal and is ignored.

Lesson 4, Q2: B. MACD line = 12 EMA − 26 EMA, so it is positive exactly when the 12 EMA is above the 26 EMA.

Lesson 4, Q3: B. The 12/26 EMA cross lags the turn substantially. The better entry is usually the first pullback after the regime changes.

Lesson 4, Q4: B. Structure had weakened: a fast, deep decline and a lower high. The filter lags structure, and structure wins.

Lesson 4, Q5: C. In a range the 12 and 26 EMAs wind around each other, so the MACD line keeps crossing zero and the filter stops being informative.

Lesson 5, Q1: B. Red bars above a positive MACD line mean speed is below its recent average. The trend can keep rising while decelerating.

Lesson 5, Q2: A. The hook is the first contraction after the pullback's bars have been growing — an early sign the counter-move is losing force.

Lesson 5, Q3: B. The hook only shows slowing. A close above the prior candle's high is evidence that price has actually turned.

Lesson 5, Q4: B. Hook entry at 111 versus the crossover close about 31 pips higher — which also allowed a much tighter stop.

Lesson 5, Q5: B. Shrinking pushes signal fading acceleration. Structure is still bullish, so the right response is defensive, not reversal.

Lesson 5, Q6: B. Twelve pips per candle against seven: the counter-move is more forceful than the trend. That is the profile of a pullback that may become a reversal.

Lesson 6, Q1: B. Bearish divergence is price making a higher high while the MACD line makes a lower high.

Lesson 6, Q2: B. The MACD line is a speed reading. Lower peaks mean the later push covered less distance per candle — nothing more.

Lesson 6, Q3: C. The rule requires the divergence, a lower high, and a close through the swing low between — structure confirms the warning.

Lesson 6, Q4: B. Without structural change, the rule never triggers, no matter how many divergences appear.

Lesson 6, Q5: A. The higher low is the structure the trade depends on. If it breaks, the reversal idea is wrong.

Lesson 6, Q6: B. Mismatched peaks produce divergences that are not real. Compare the indicator at the same candles as the price swings.

Lesson 7, Q1: B. The reset is speed being spent during the pullback while the 12 EMA stays above the 26 EMA.

Lesson 7, Q2: B. A close beyond the 50 EMA cancels the setup. A strong bounce does not re-qualify it.

Lesson 7, Q3: B. The reset rule requires the MACD line to fall below half its impulse peak while staying above zero.

Lesson 7, Q4: B. The pullback low is the structure the trade depends on. If price breaks it, the pullback has become something else.

Lesson 7, Q5: B. If the nearest obstacle gives less than 1.5R, the reward does not justify the risk. Moving the stop to fix the maths is not allowed.

Lesson 7, Q6: B. Every method has valid losing trades. Fixed, small risk keeps them affordable. Rules are judged over many trades, not one.

Lesson 8, Q1: B. Each crossover needs several candles to confirm; in a range that is about half the swing, so buys print mid-rally and sells mid-decline.

Lesson 8, Q2: B. Shrinking MACD swings usually appear first, before the 50 EMA has fully flattened and before many zero crosses have accumulated.

Lesson 8, Q3: B. Range symptoms are present. The reset method depends on a trend impulse, which does not exist here.

Lesson 8, Q4: B. The range floor is the structure the trade depends on. If it breaks, the range has ended.

Lesson 8, Q5: B. Leaving the narrow band around zero decisively is the first sign of a new trend. Range methods should be switched off.

Lesson 8, Q6: B. The flat average and repeated highs and lows showed a range from early on — the cue to switch crossover trading off.

Lesson 9, Q1: B. The regime decides which rule set applies. Entries found before classification are often the right method in the wrong market.

Lesson 9, Q2: B. Direction comes from the higher timeframe. An H1 uptrend inside an H4 downtrend is a counter-trend rally.

Lesson 9, Q3: D. Plain crossovers have no structural stop and whipsaw in ranges. They survive only as ideas inside other methods.

Lesson 9, Q4: B. Compounded: 1 − 0.99^8 ≈ 7.7%. At 3% per trade the same streak costs about 22%.

Lesson 9, Q5: B. Divergence is a reason to protect, not to reverse. Structure decides the exit.

Lesson 9, Q6: B. 0.4 × 2 − 0.6 × 1 = 0.8 − 0.6 = +0.2R per trade. The figures are hypothetical, used to show the arithmetic.

Lesson 10, Q1: B. Direction comes first from the higher timeframe. Everything else is filtered through it.

Lesson 10, Q2: B. The method uses the hook and a price trigger; the later crossover would have meant a worse entry and a wider stop.

Lesson 10, Q3: C. Rule set B requires divergence, then a lower high, then a close through the swing low between.

Lesson 10, Q4: C. A valid loss is part of any method. Changing rules after one trade, or sizing up to recover, replaces process with emotion.

Lesson 10, Q5: B. Ranges defeat MACD's trend methods. The passport lists them, counter-trend crossovers, unconfirmed divergence and higher-timeframe conflict under 'avoid'.

Lesson 10, Q6: B. Taking a trade that breaks the alignment rule is a process error, regardless of outcome.